



Controller Debugging Manual

2023.05.01

A decorative graphic on the left side of the page. It features a large brown circle with the word 'contents' inside. Above it is a smaller light beige circle with a right-pointing arrow. To the left is a small dark brown circle with a left-pointing arrow. Below the brown circle is a small orange circle with a right-pointing arrow. Several thin white lines radiate from the central circles, connecting to other circles of various sizes (orange, beige, brown) scattered across the left side of the page.

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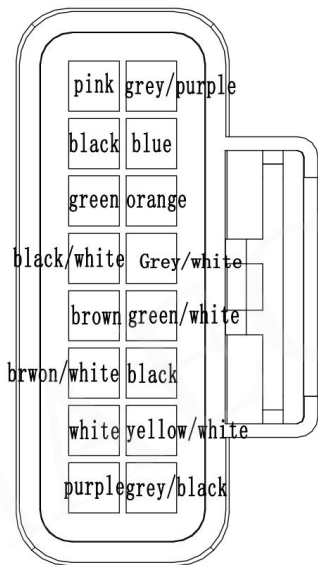
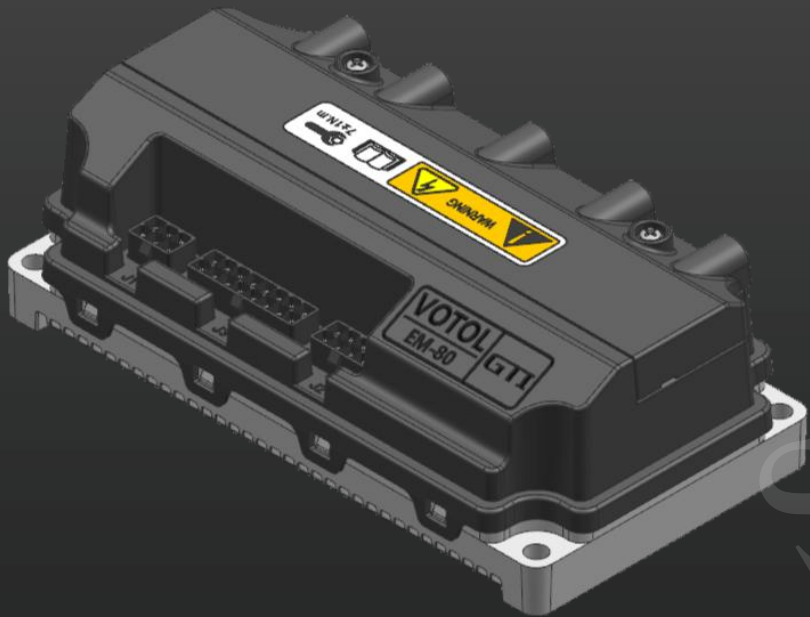
04

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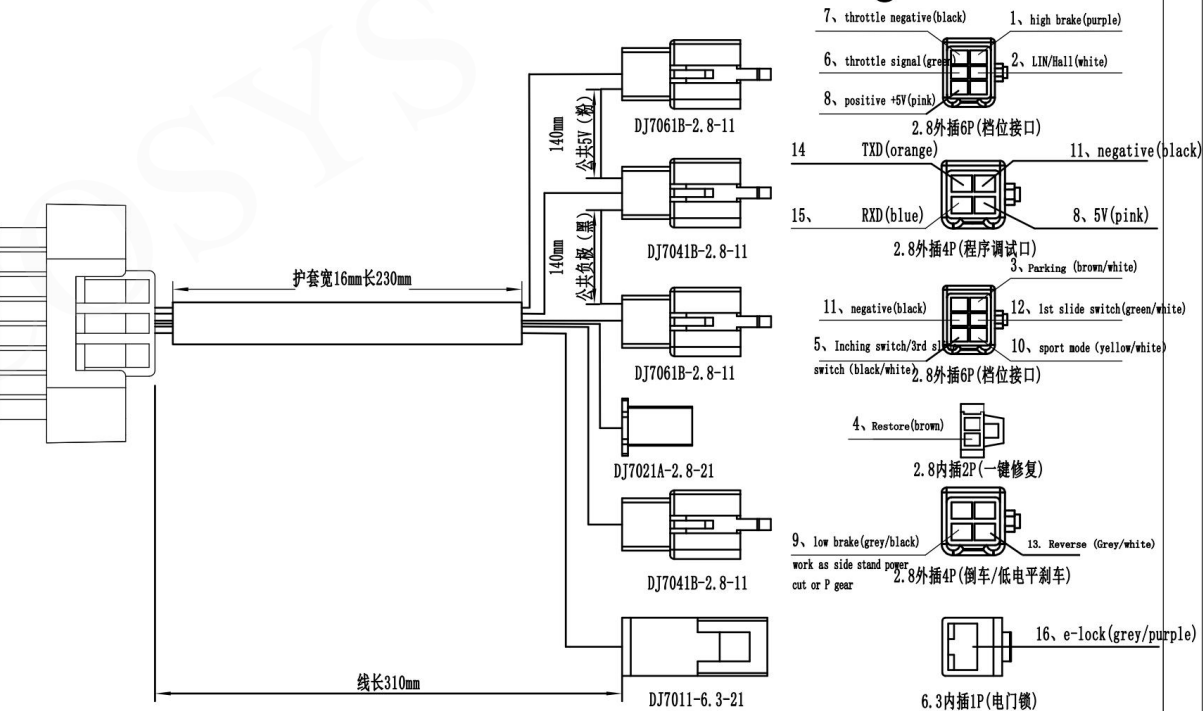


5557-2-2*8公胶壳（功能接口）

技术要求

- 1、所有电线线径采用0.3mm²。
- 2、所有插件采用镀磷铜插件；所有插件采用“振特”品牌。
- 3、线长按照图纸标示。
- 4、塑料插件为增强尼龙。
- 5、线头拉脱力要求45~60N之间。
- 6、端子与公胶壳拉脱力要求60N以上。
- 7、所有5557插头端线束必须打胶。
- 8、所以5557插头端必须放置专用防水硅胶垫。

SIAECOSYS 2x8 main function wiring



5557-2（2*8主线缆）

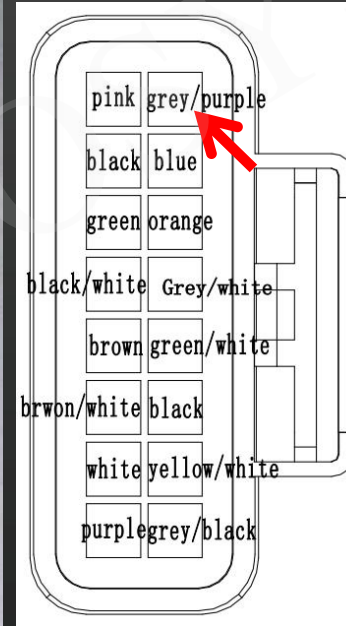
客户:	技术:	
	生产:	日期: 2020.05.07
客户编号:	质检:	单位: mm
备注:		
客户单位确定 签字盖章 (销售部)		

Wiring instructions

1.1 Function wires

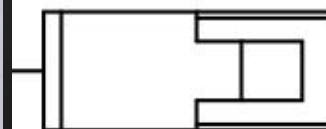
E-lock connector (grey/purple)

Connecting with ignition key

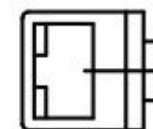


E-lock
(grey/purple)

Connecting with
ignition key



DJ7011-6. 3-21



16、e-lock(grey/purple)

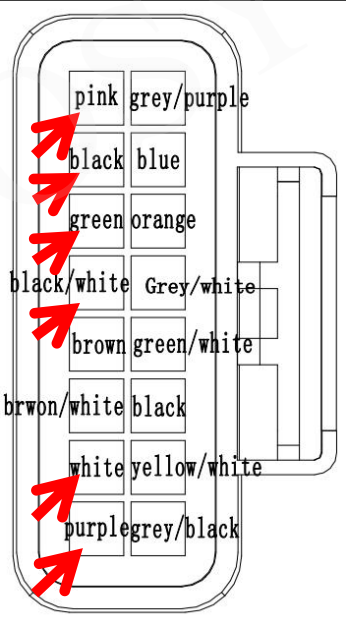
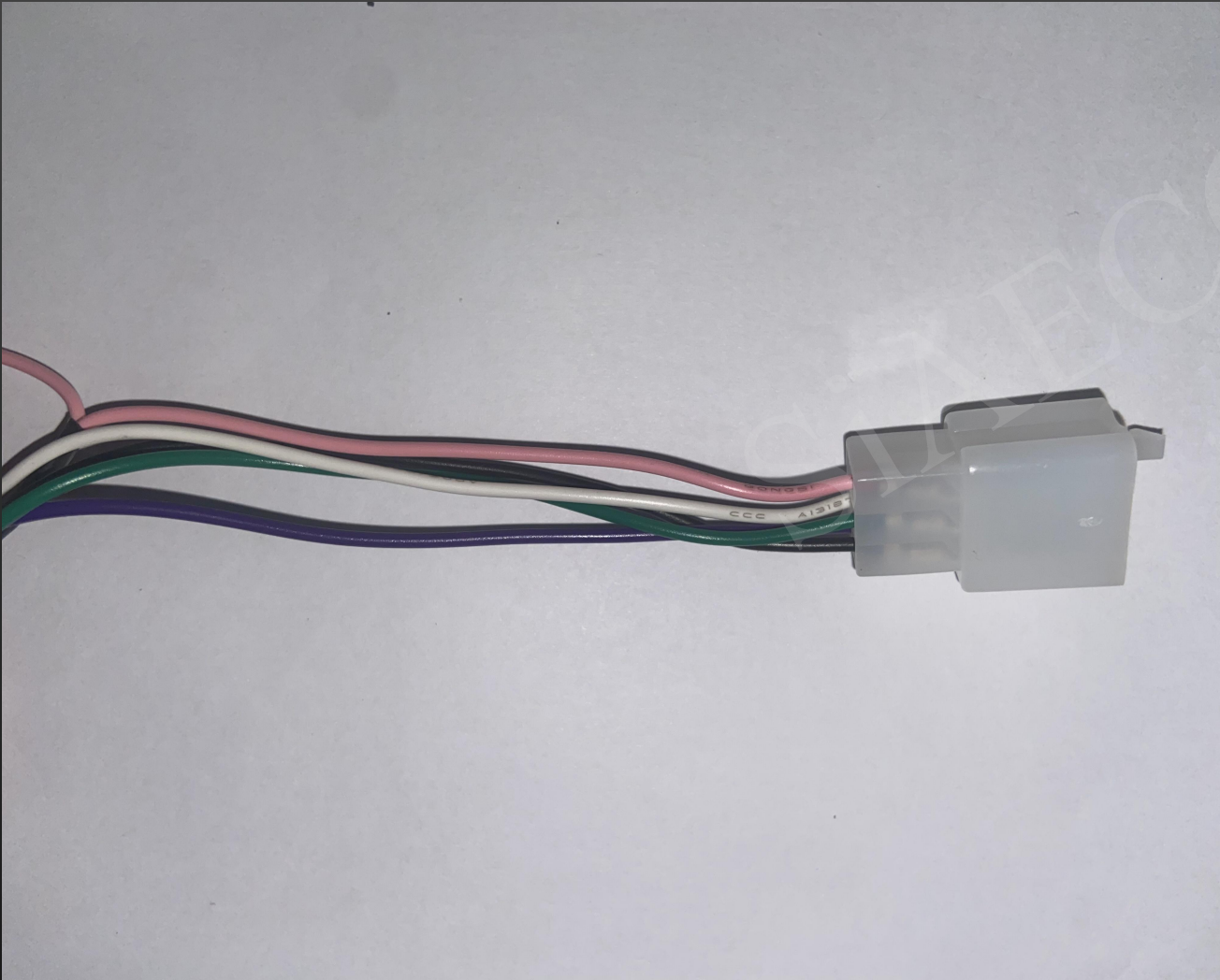
6. 3内插1P(电门锁)

Wiring instructions

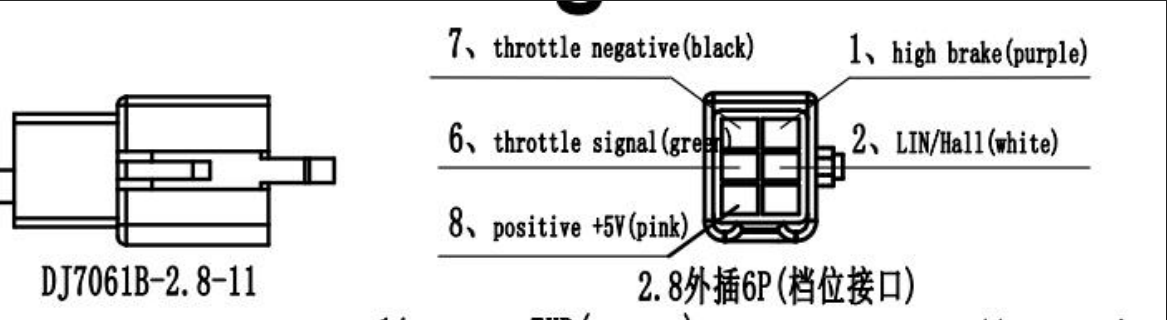
1.1Function wires

Throttle connector

Connecting with throttle, display and brake



Throttle 5V+ (Pink)	Connecting with throttle
Throttle - (Black)	
Throttle signal (Green)	
Display signal (White)	Connecting with Display
Brake signal (Purple)	Connecting with brake (12V)

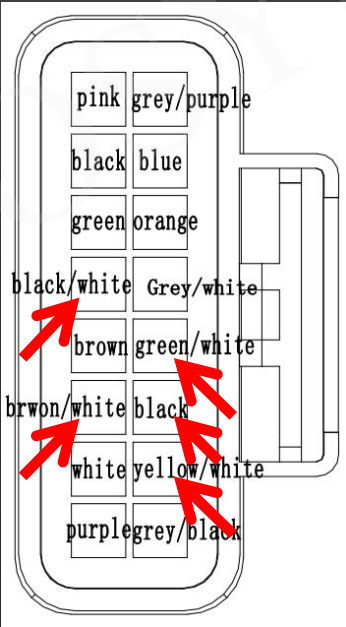


Wiring instructions

1.1Function wires

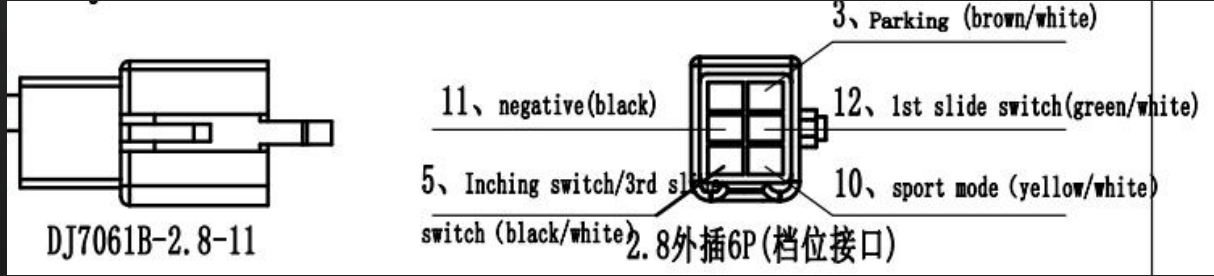
Gears connector

Connecting with gears switch



Gear 3 (Black/white)
Negative - (Black)
Gear 1 (Green/white)
Gear Sport (Yellow/white)
Gear Park (Brown/white)

**Connecting with
gears switch, and
the negative wire
must be
connected**

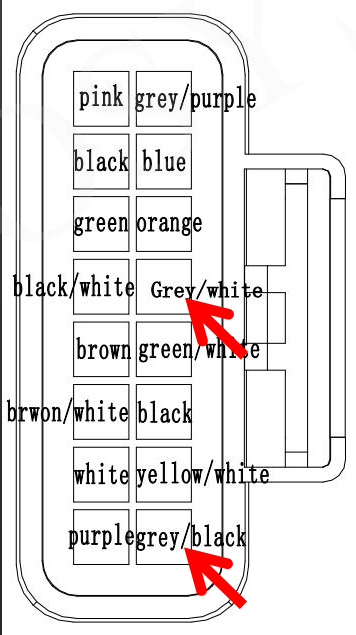


Wiring instructions

1.1Function wires

Back & kick stand connector

Connecting with back switch and kick stand switch



Back (Grey/white)	1、 Connecting the reverse switch with GND 2、 Press the button and apply the throttle, the motor will in reversing rotation
Kick stand (Grey/black)	1、 Connecting the kick stand switch with GND



Wiring instructions

1.1 Function wires

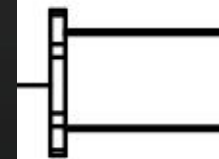
Restore connector

Connecting with restore button



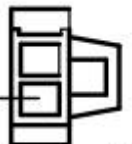
Restore
(Brown)

1、
When the
throttle and
brake errors
appear, double-
press the
restore button
and with long
press at second
time to active
resote function



DJ7021A-2.8-21

4、 Restore(brown)



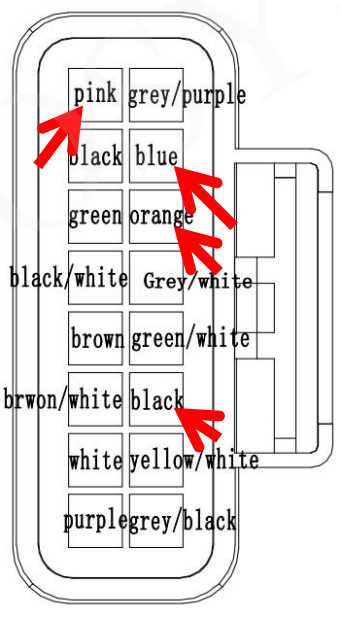
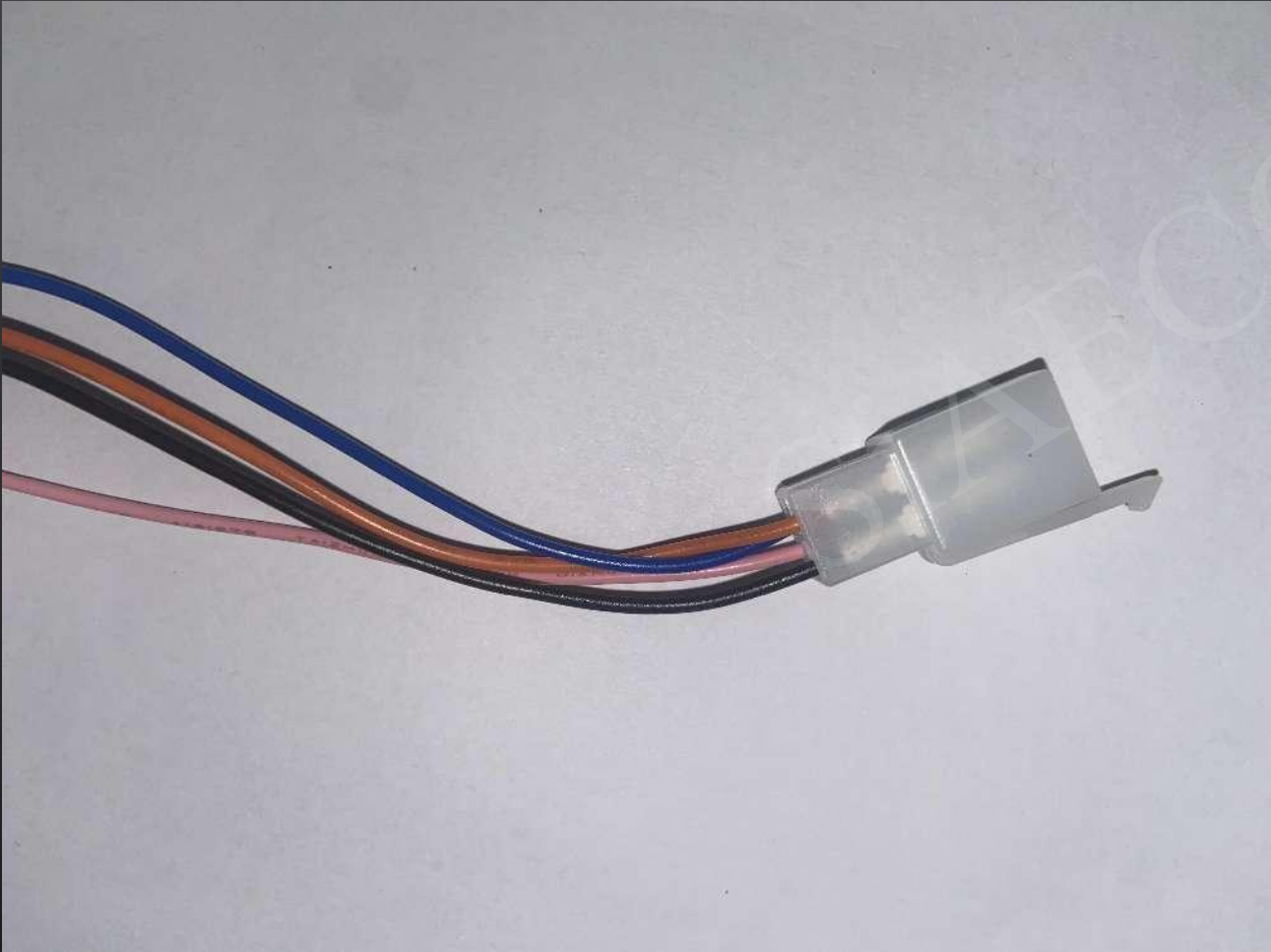
2.8内插2P(一键修复)

Wiring instructions

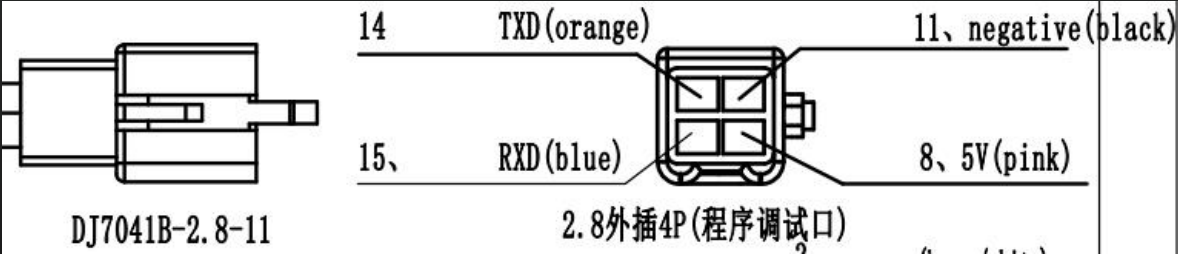
1.1Function wires

Debugging connector

Connecting with PC USB cable and bluetooth

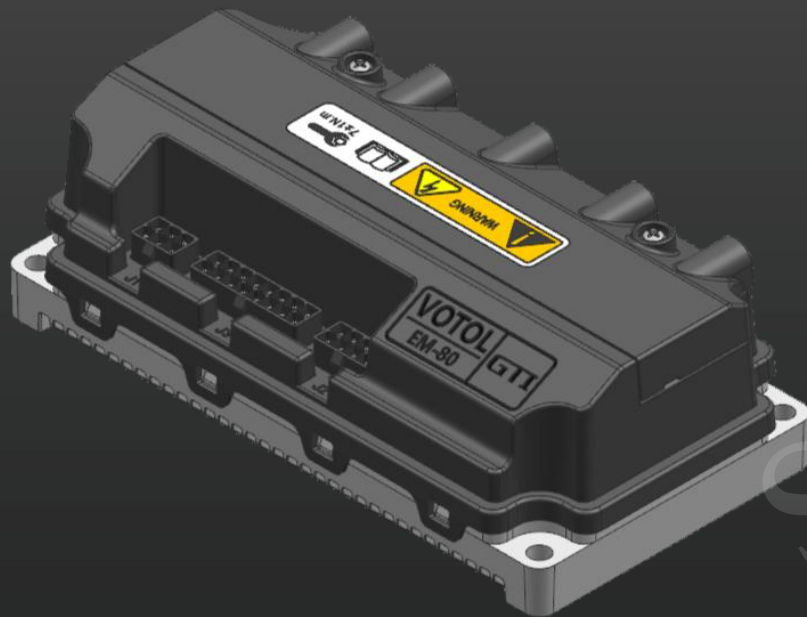


5V (Pink)	1、Connecting with programmed USB cable and bluetooth (Don't plug and unplug when the controller is powered on)
Negative pole (Black)	
Debugging RXD (Blue)	
Debugging TXD (Orange)	

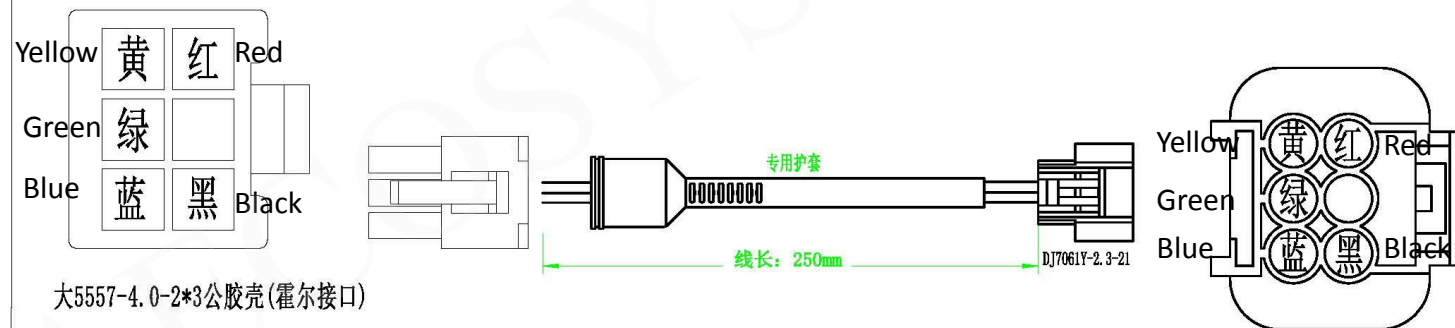


Wiring instructions

1.2 HAll Connector



蓝德电子科技有限公司(事业三部)



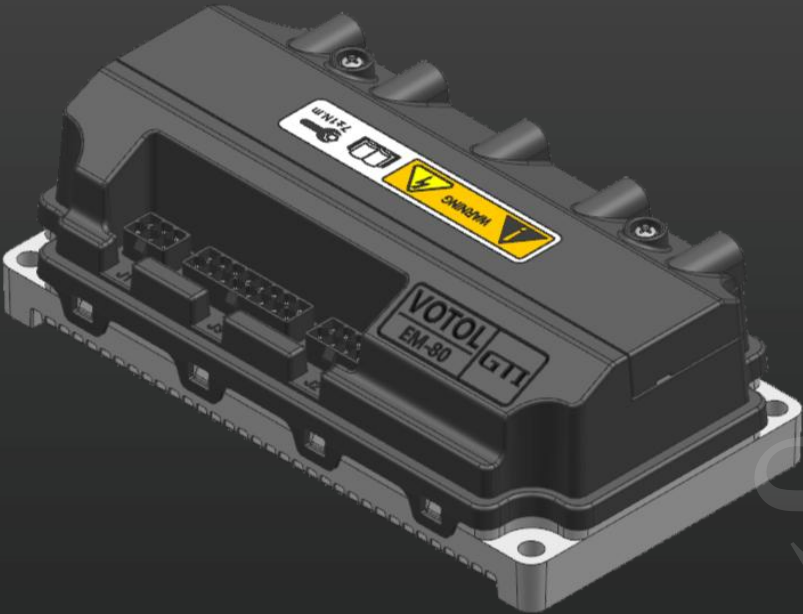
技术要求

- 1、所有电线线径采用0.3mm²。
- 2、所有插件采用镀磷铜插件；
所有插件采用“振特”品牌。
- 3、线长按照图纸标示。
- 4、塑料插件为增强尼龙。
- 5、线头拉脱力要求45~60N之间。
- 6、端子与公壳拉脱力要求60N以上。
- 7、所有5557插头端线束必须打胶。

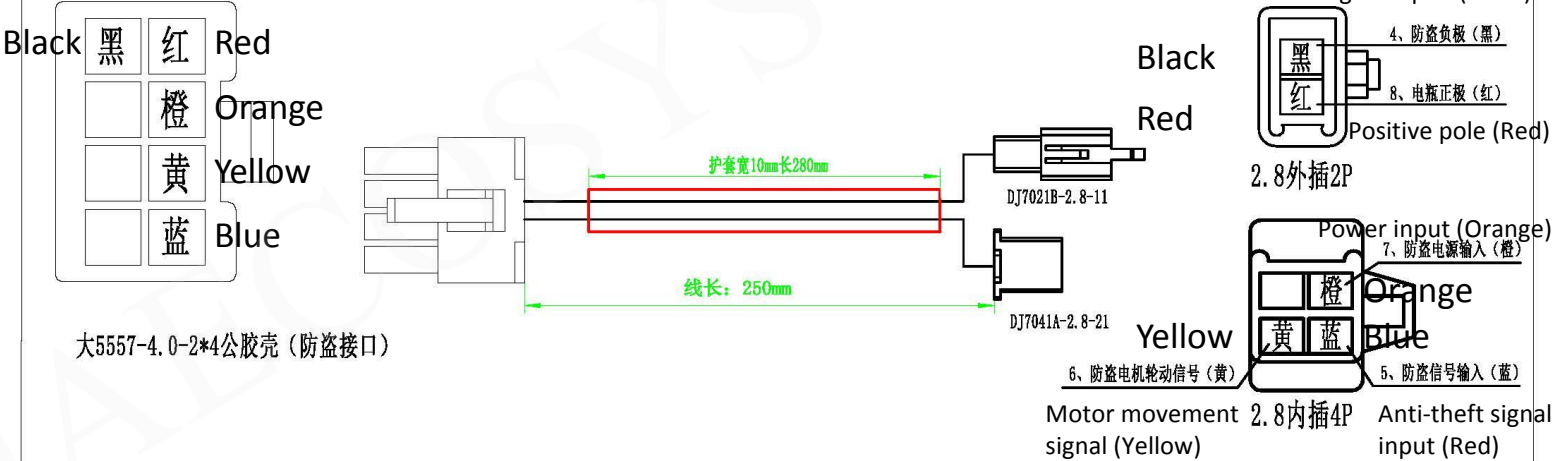
VOTOL 蓝德科技		大5557劲浪转接线(2*3霍尔转接线)	
客户: 劲浪	技术:		
	生产:	日期: 2020. 09. 09	
客户编号:	质检:	单位: mm	
备注:			
客户单位确定 签字盖章 (销售部)			

Wiring instructions

1.3 Anti-theft connector



蓝德电子科技有限公司(事业三部)



技术要求

- 1、所有电线线径采用0.3mm²。
- 2、所有插件采用普通插件。
- 3、线长按照图纸标示。
- 4、塑料插件为增强尼龙。
- 5、线头拉脱力要求30~60N之间。
- 6、端子与公壳拉脱力要求65~80N之间。
- 7、所有5557插头端线束必须打胶。

VOTOL 蓝德科技 大5557乖乖兔转接线- (2*4防盗器)		
客户:乖乖兔	技术:	
	生产:	日期: 2020.09.09
客户编号:	质检:	单位: mm
备注:		
客户单位确定 签字盖章 (销售部)		

Software connection manual

2.1 Software installation

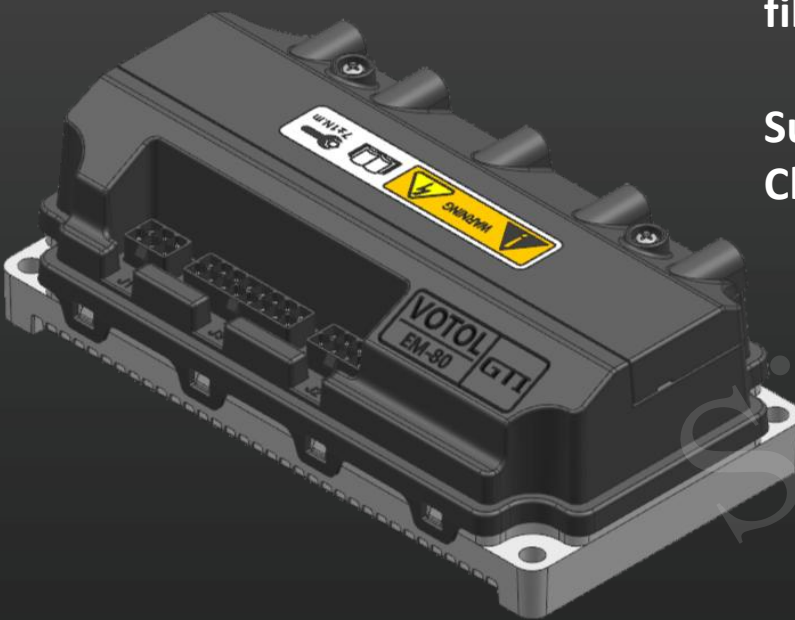
Connecting the controller to the PC with the USB debugging cable that comes with the controller, the driver will be installed.

If failed, please follow the below steps:

Decompress the file “ USB-to-serial-win10-20150814 ”, then decompress the file “ YH-340 USB ”.

Supported system: WIN7/WIN8/WIN10

Choosing and decompressing the correct file based on your PC system.



windows 7 8 10 32_64	2019-08-07 13:52	文件夹	
win_me_2000_XP USB-to-Serial	2013-03-08 14:38	360压缩 ZIP 文件	1,586 KB
YH-340 USB	2013-03-08 14:37	360压缩 ZIP 文件	295 KB
简介	2015-08-07 10:56	文本文档	2 KB

1. Decompression and start

Setup 32.64位元

2012/8/3 12:31

应用程序

3,075 KB

2. Installing according to the above steps

After installation, connecting the USB cable with PC.

If failed, please right click My PC and enter in the PC management interface.



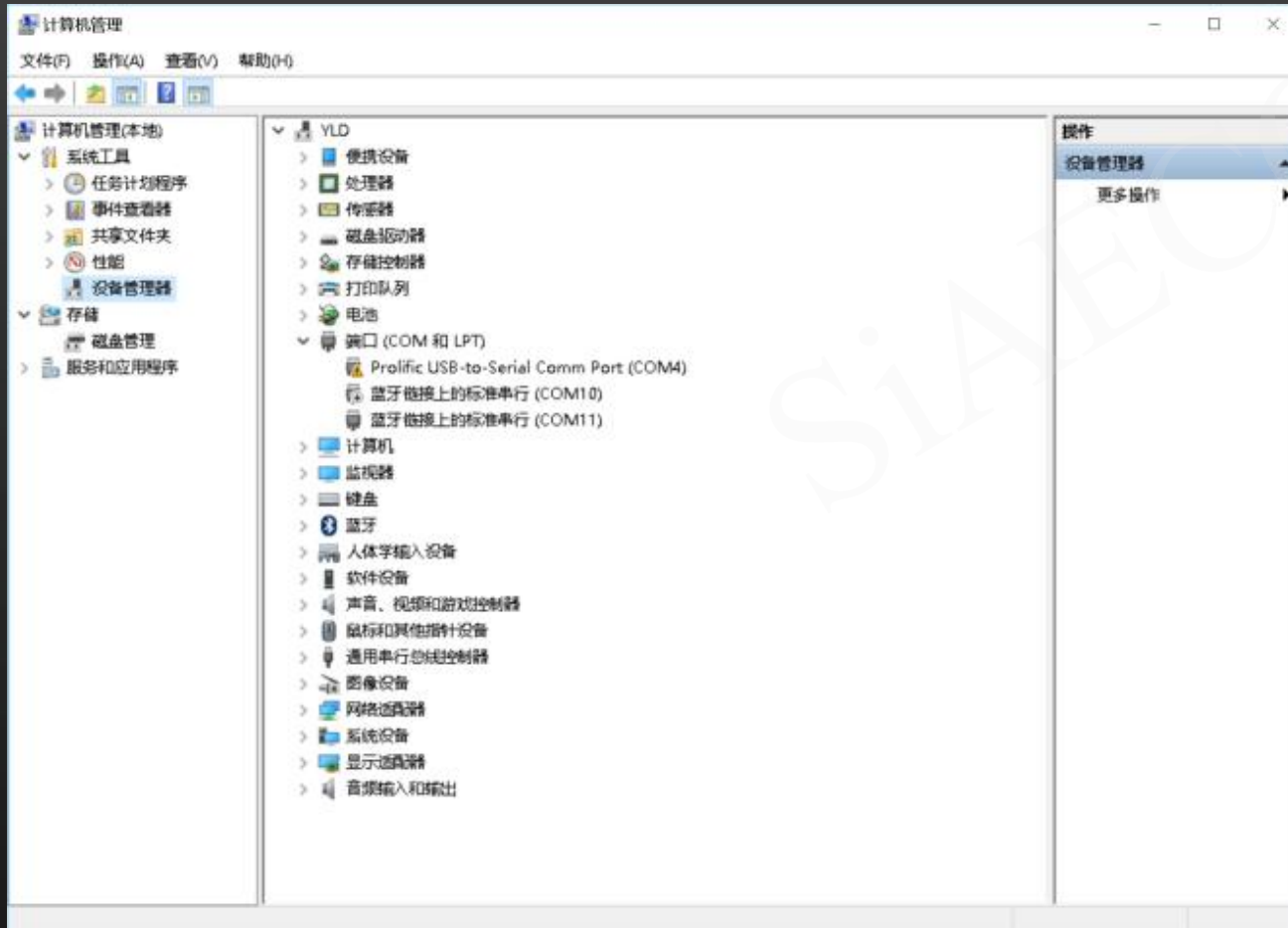
Software connection manual

2.1 Software installation

Clicking “Device manager”

Clicking “COM and LPT”

Right clicking the port with “!” and choosing “Update the driver software”



Software connection manual

2.1 Software installation

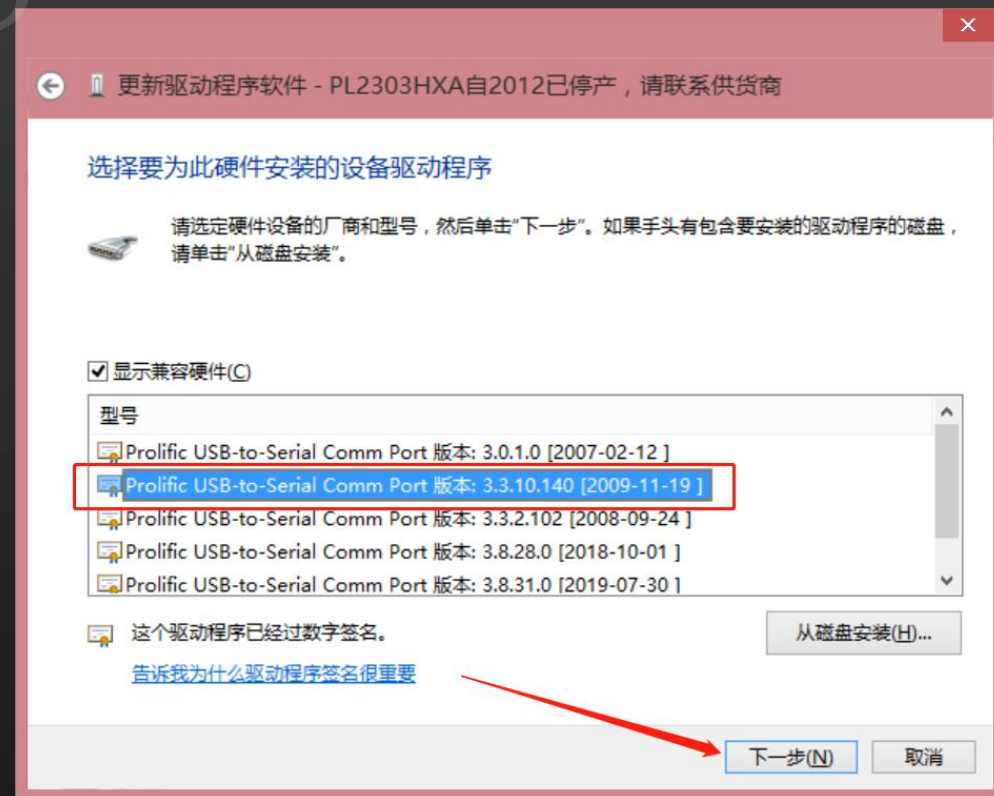
If the port shows “PL2303HXA自2012已停产”,

Step 1: Right clicking and choosing driver update;

Step 2: clicking browse my computer for driver

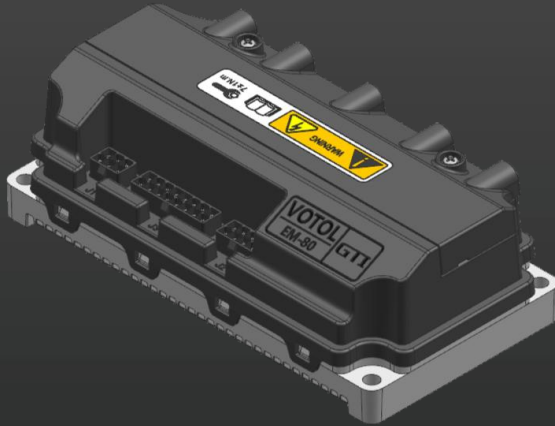
Step 3: Choosing 3.3.10.140 (2009-11-19) version, and clicking next step.

Closing the interface when finished, and start the controller software to connecting the controller with PC



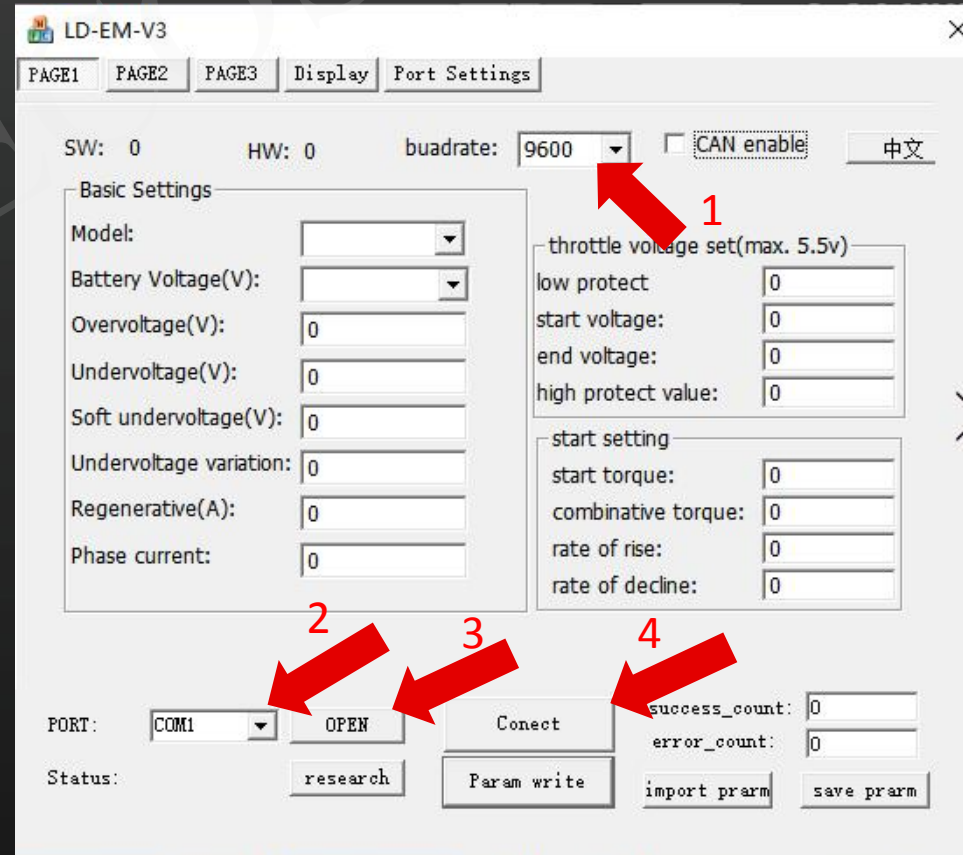
Software connection manual

2.2 Connecting debugging cable with controller and PC



Step 1: Selecting the correct USB debugging cable according to the controller with or without CAN function;

Step 2: Choosing **9600 for buadrate**, and opening the device manager to check whether the USB port is installed successfully before clicking "connect". If it's correct, **selecting and starting the proper PC controller software**, then connecting the **correct port** when the **controller is power on**. All parameters will be showed in the software.



Software connection manual

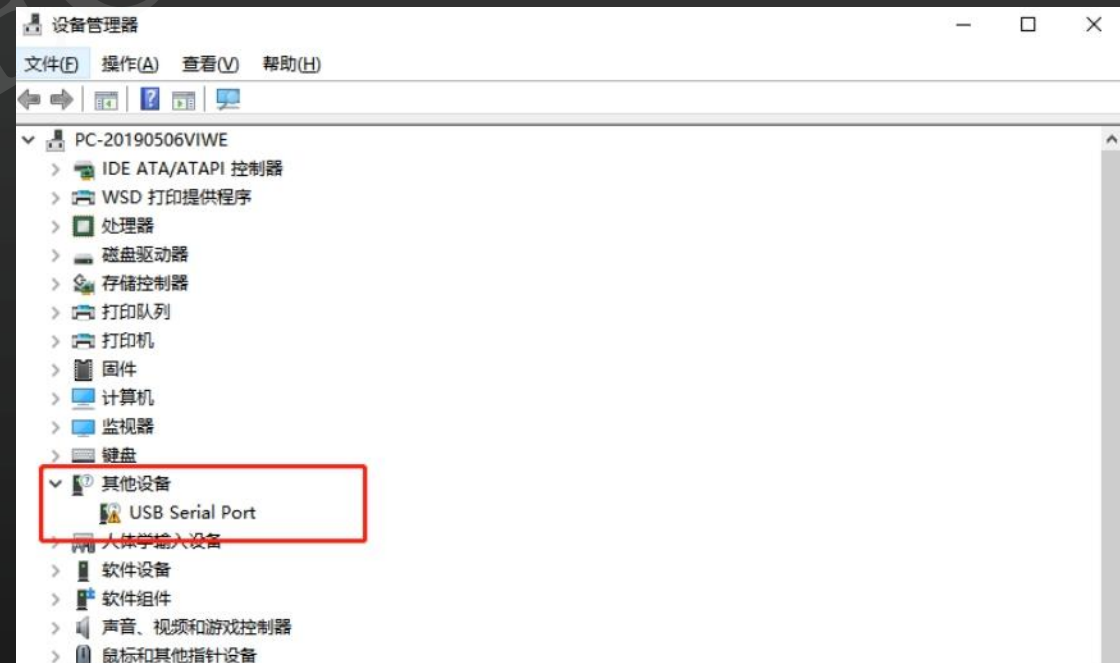
2.3 Installation of debugging cable with CAN function



Step 1: Connecting the USB cable with PC, and it will be installed of itself when the PC internet is available. If not, decompressing the ft232r driver



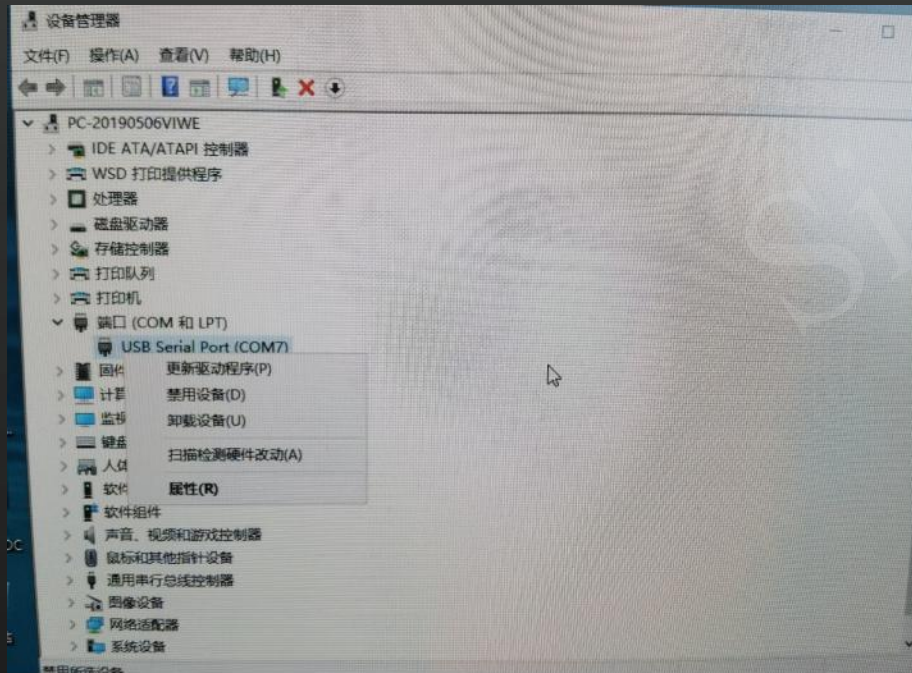
Step 2: Connecting the USB cable with PC and opening the device manager. If the port with “!” like below picture, the driver should be installed.



Software connection manual

2.3 Installation of debugging cable with CAN function

Step 3 : Right-clicking the USB Serial like below picture, then choosing soft update.



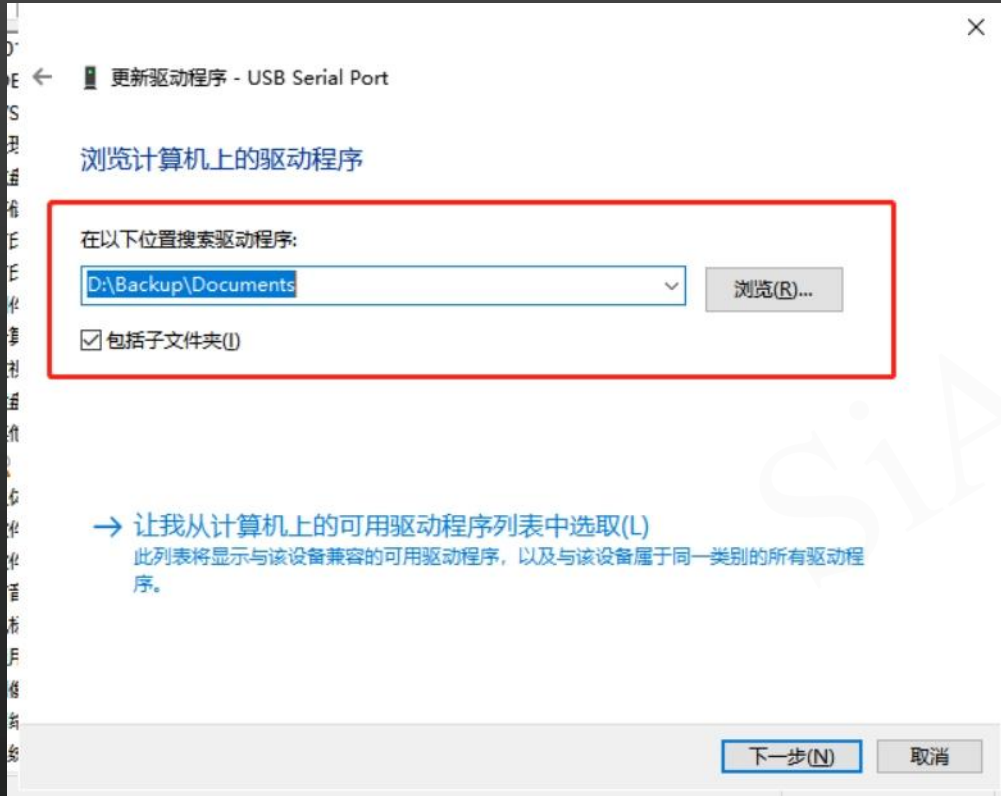
Step 4: Clicking browse my computer for driver like below picture.



Software connection manual

2.3 Installation of debugging cable with CAN function

Step 5: Selecting the correct the driver in the file like below picture, then clicking the next step for the driver installation

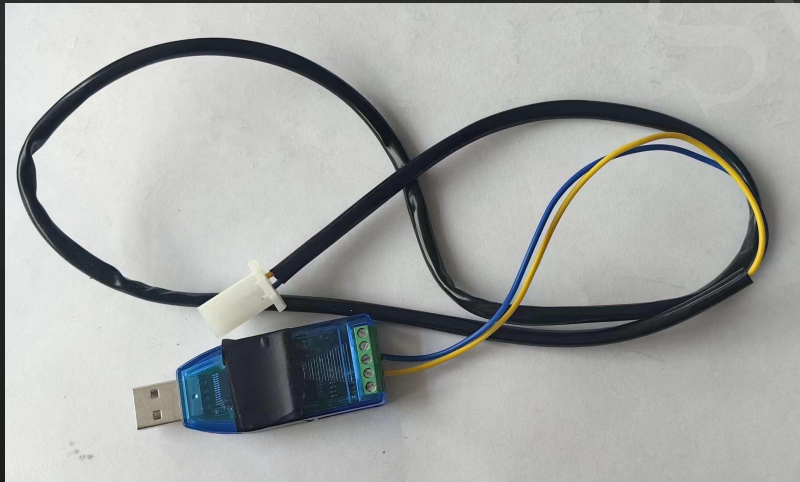


Step 6: When the port showing up, it means the driver is installed successfully;
If the “!” is still there, please reconnect the USB debugging cable.



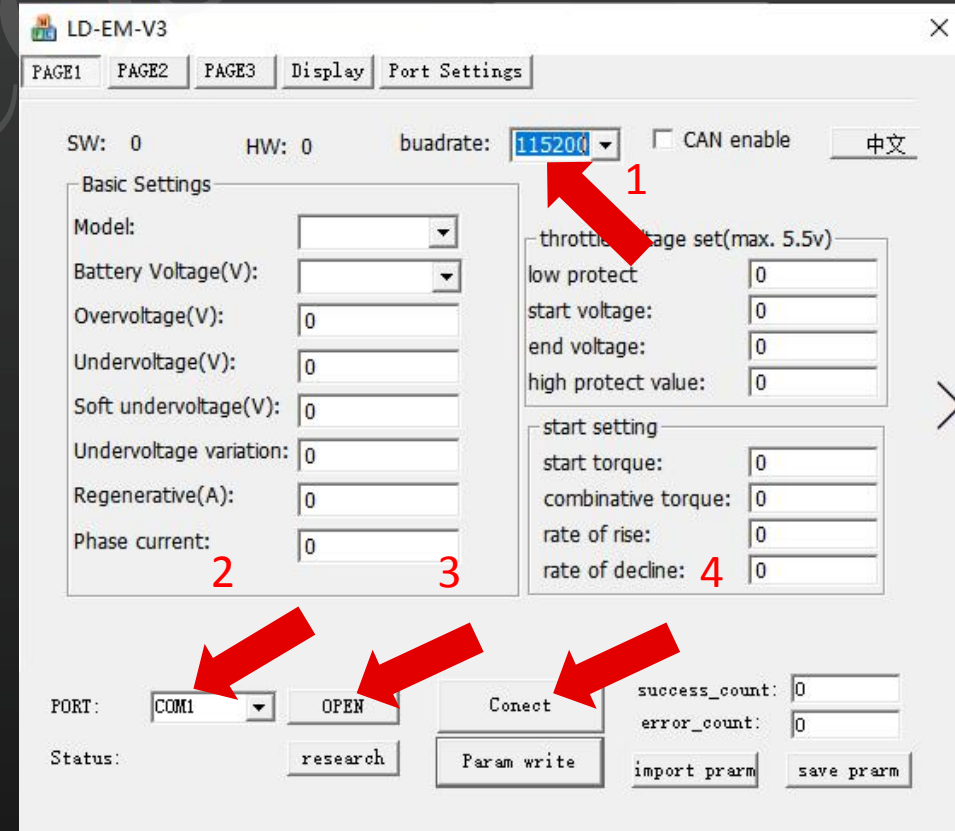
Software connection manual

2.4 Installation of debugging cable with CAN function



Step 1: Selecting the correct USB debugging cable according to the controller with or without CAN function;

Step 2: Choosing **115200 for baudrate**, and opening the device manager to check whether the USB port is installed successfully before clicking “connect”. If it’s correct, **selecting and starting the proper PC controller software**, then connecting the **correct port** when the **controller is power on**. All parameters will be showed in the software.



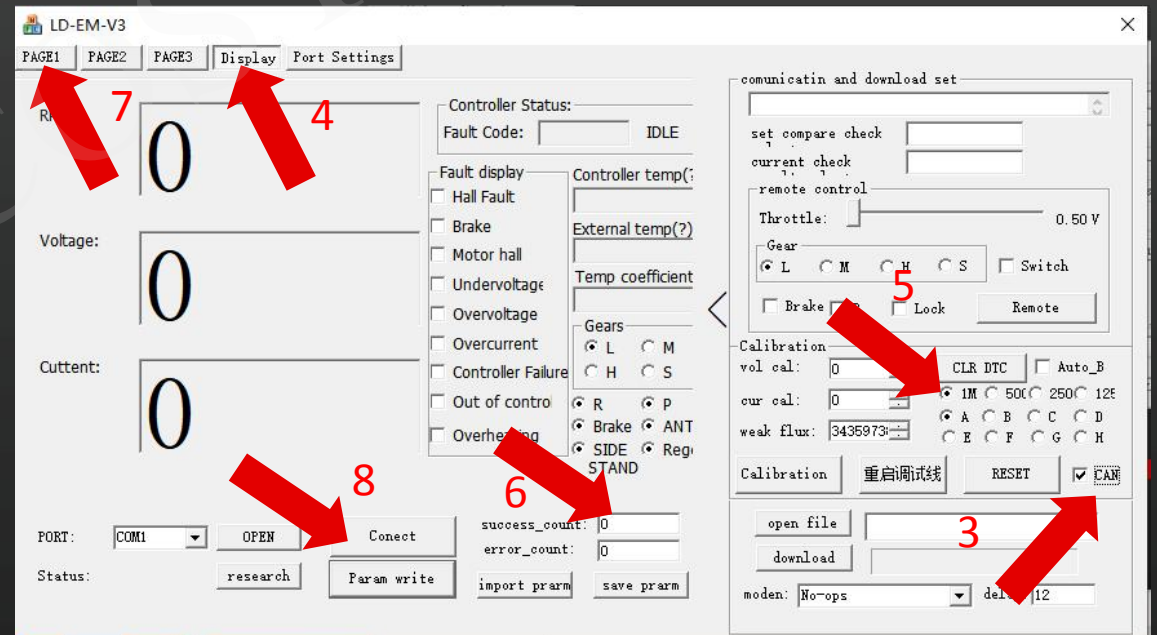
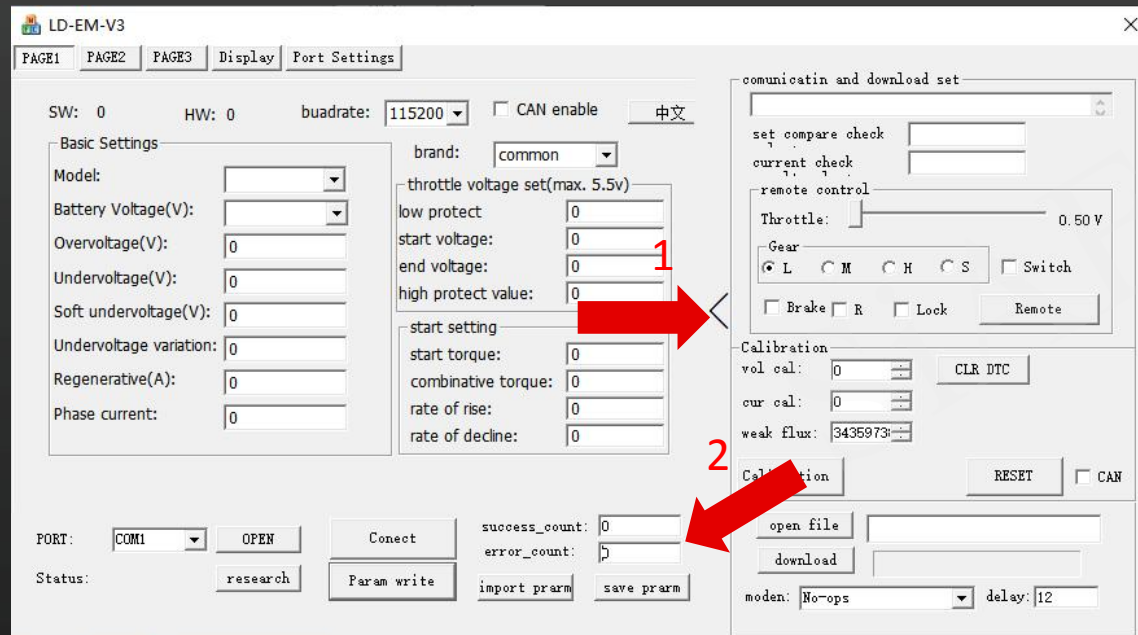
Software connection manual

2.4 Installation of debugging cable with CAN function

Step 3: If the controller can't be connected, please enter the advanced interface and choose the CAN with correct ID or Auto_B

Following the below steps: Entering the advanced interface and click display interface, choosing the correct CAN baudrate and ID based on the controller model, meanwhile, observing the the value, when it's rised, click page 1 to connect the controller

success_count:	0
error_count:	0



Type "~" and press "shift" to enter advanced interface

Parameter setting description

3.1 Parameter page 1

The screenshot shows the 'LD-EM-V3' software window with the 'Port Settings' tab selected. The interface includes fields for 'SW: 0', 'HW: 0', and a 'buadrate:' dropdown menu set to '115200'. A 'CAN enable' checkbox is present, along with a language switch button labeled '中文'. The 'Basic Settings' section contains various input fields for Model, Battery Voltage, Overvoltage, Undervoltage, Soft undervoltage, Undervoltage variation, Regenerative current, and Phase current. The 'throttle voltage set(max. 5.5v)' section includes fields for low protect, start voltage, end voltage, and high protect value. The 'start setting' section includes fields for start torque, combinative torque, rate of rise, and rate of decline. At the bottom, there are buttons for 'PORT: COM1', 'OPEN', 'Connect', 'Status:', 'research', 'Param write', 'import prarm', and 'save prarm'. A 'success_count' and 'error_count' field are also visible.

Annotations with arrows point to the following elements:

- Language switch (Chinese - English)
- CAN buadrate
- Without CAN: 9600
- With CAN: 11520
- Connecting the USB cable, choosing the correct port and click open. Clicking connect to read the controller parameters when the status shows connected. If the port not shows up, please click research.
- Click param write after adjusting the parameters.
- Import the settings that saved on the computer.
- Save the prarm on the computer.

A close-up of the 'buadrate:' dropdown menu showing the following options: 115200, 9600, 115200, and 256000. The '115200' option is currently selected.

language switch
(Chinese - English)

CAN buadrate
Without CAN: 9600
With CAN: 11520

Connecting the USB cable, choosing the correct port and click open. Clicking connect to read the controller parameters when the status shows connected. If the port not shows up, please click research.

Click param write after adjusting the parameters.

Import the settings that saved on the computer.

Save the prarm on the computer.

Parameter setting description

3.1 Parameter page 1

SW: 0 HW: 0 buadrate: 9

Basic Settings

Model:

Battery Voltage(V):

Overvoltage(V):

Undervoltage(V):

Soft undervoltage(V):

Undervoltage variation:

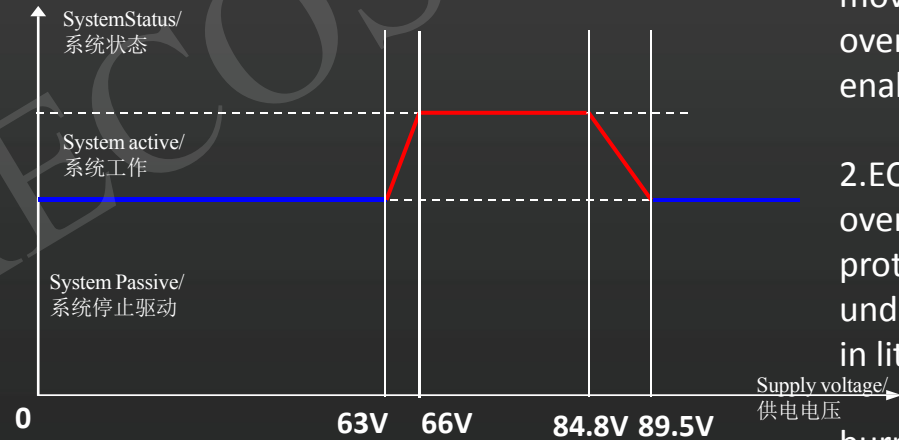
Regenerative(A):

Phase current:

Selecting the controller working voltage range. Software will preset lead-acid battery's overvoltage, undervoltage, soft undervoltage parameters (lithium batteries need to be filled in more actual conditions)

Battery Voltage(V):	
Overvoltage(V):	48V
Undervoltage(V):	60V
Soft undervoltage(V):	72V
	84V
	96V

72V power supply



Attention:

1. Overvoltage error: The vehicle is not moved after opening e-lock. The overvoltage protection function will be enabled.
2. ECU undervoltage error: Due to the overvoltage of the lithium battery protection board is set too high or the undervoltage setting is too low, resulting in lithium battery protection and MOS

Undervoltage variation is normally set to 1

After the undervoltage fault occurs, the voltage of the vehicle can be recovered only when it is higher than the undervoltage value of 1V

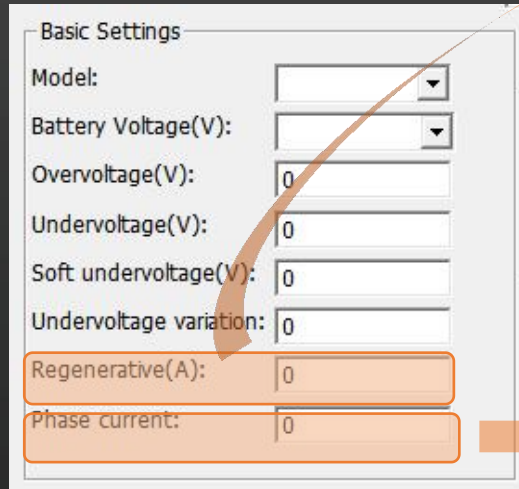
For example:

When the undervoltage is set to 63V, the vehicle battery voltage is 62V when start charging, and the vehicle can not drive until the battery is charged to 63V.

- 1、 Default basic value of lead-acid battery's overvoltage, undervoltage, and soft undervoltage : Due to software deviation range, the original voltage compensation is +1V.
- 2、 The lithium battery parameters are set according to battery. Due to the software deviation range, the original voltage compensation is +1V. and the original lithium battery parameters are compensated by +1V, so for a total compensation of +2V.
- 3、 The soft undervoltage value is set by 3V higher than undervoltage 's. When the voltage value is lower than the soft undervoltage value, the controller output current and speed will be reduced to protect the battery and increase the battery life before undervoltage.

Parameter setting description

3.1 Parameter page 1



Basic Settings	
Model:	
Battery Voltage(V):	
Overvoltage(V):	0
Undervoltage(V):	0
Soft undervoltage(V):	0
Undervoltage variation:	0
Regenerative(A):	0
Phase current:	0

Regenerative current

1、 The regenerative current should be set according to the maximum reverse charge current that battery can take.

If the battery does not support reverse charging, set the value to 0.

Cautions:

1、 If the battery does not support reverse charging, or the regenerative current is set too high, it will cause the battery BMS protection due to the high reverse charging current when losing throttle or braking. (Notice: Changing battery, please turn off the regenerative function)

Phase current:

1、 Different models of controllers get different max phase current value, please set according to the value that provided by tech department. (For details, refer to the technical department document: Capacitor Protection Settings)

2、 The 100% means the max value of the phase current. When limiting the phase current, you can fill in the parameter according to the proportional .

3、 The phase current limit can only be less than or equal to the parameter provided by the technical department.

Cautions:

1、 If the phase current limit is set too high, the temperature of the controller will increase sharply which will burn out the MOS tube.

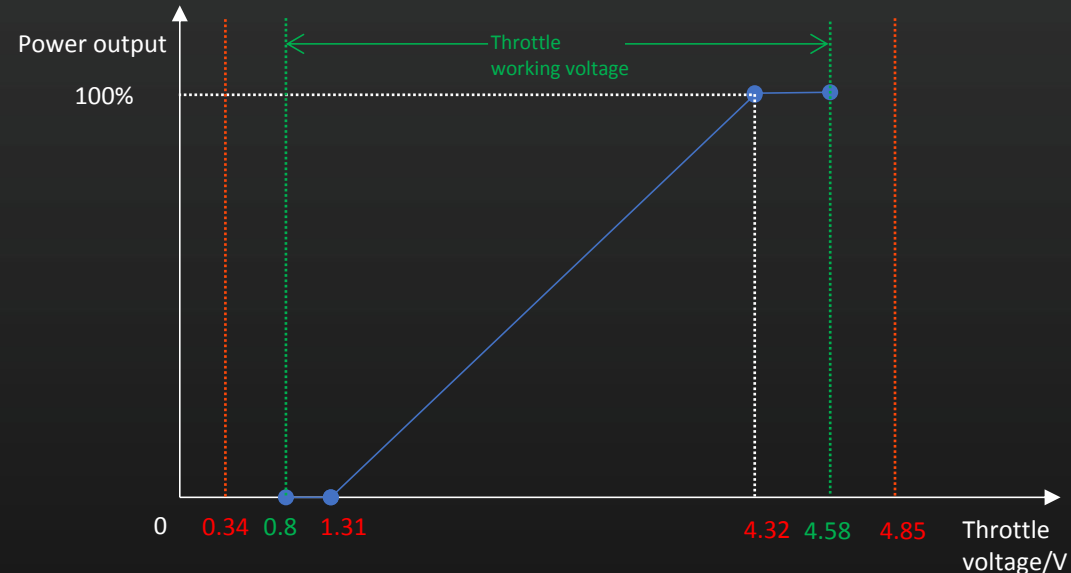
Parameter setting description

3.1 Parameter page 1

throttle voltage set(max. 5.5v)	
low protect	0
start voltage:	0
end voltage:	0
high protect value:	0



1. Throttle voltage set: Set the start and end voltage according the **throttle actual lowest and highest voltage**.
2. Throttle low protect: When the throttle signal voltage is lower than this value during power-on, throttle error will occur.
3. Throttle start voltage: the minimum level of valid speed, If the throttle signal voltage is higher than this value, the motor starts to run, otherwise, the motor stops.
4. Throttle end voltage: the maximum level of valid speed, which is the maximum speed value voltage. When throttle voltage reaches this value, the controller will be at full output.
5. Throttle high protect value: The motor will stop when the throttle voltage is higher than this value.



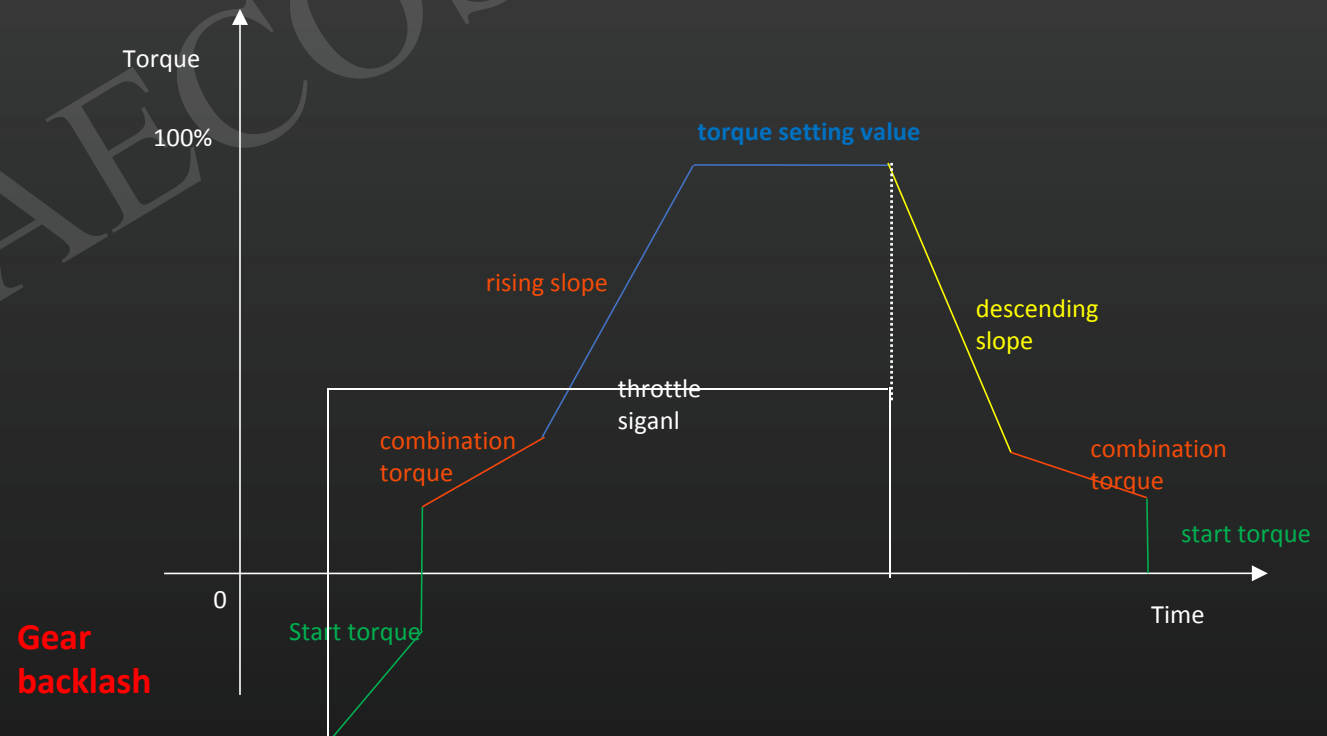
Parameter setting description

3.1 Parameter page 1

start setting	
start torque:	0
combinative torque:	0
rate of rise:	0
rate of decline:	0

1. Start torque: The start torque is usually set to 0, and no need to change.
2. Combination torque: The starting delay parameter ranges from 0 to 350. The larger the value, the more obvious the starting delay is.

(Notice: The above two settings are suitable for **mid drive motors with gearbox**, which used in the gear backlash)

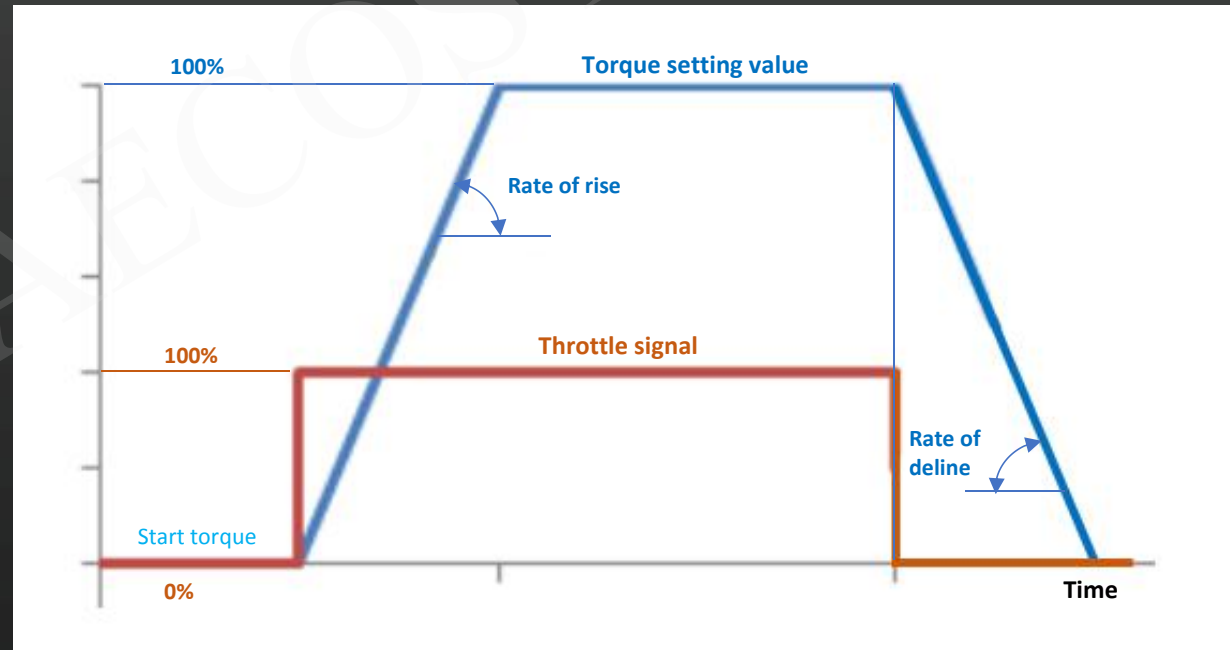


Parameter setting description

3.1 Parameter page 1

1. Rate of rise: Range 10-255. The higher the value, the faster the acceleration。
2. Rate of decline: Range 10-200. The higher the value is, the more obvious the speed drop is, and the inertia of the loose throttle decreases。

start setting	
start torque:	0
combinative torque:	0
rate of rise:	0
rate of decline:	0



Parameter setting description

3.2 Parameter page 2

Max main bus current: it should be set by the guidance of engineer

Notice: The ECU MOS will burn out if the main bus current is too high.

Max flux-weakening value:

Flux weakening control: the weak magnetic function makes the motor run above the rated speed by weakening the motor magnetic field.

Flux weakening control can broaden the speed range of permanent magnet synchronous motor. The higher the weak magnetic value, the higher the motor speed (Max value: 5000)

Notice:

1、Hub motor: The flux weakening speed of 60KM/H motor is less than 120%; the flux weakening speed of 80KM/H motor is less than 125%; the flux weakening speed of 110KM/H motor is less than 135%。

2、Inner rotor motor-Hall: The internal Hall flux weakening magnetic speed <135%; the external Hall flux weakening magnetic speed <170%

3、Inner rotor motor-Encoder: The internal Encoder flux weakening magnetic speed <230%.

4、High flux weakening value can easily lead to demagnetization of the motor and controller MOS burning

Current/Flux-Weakening

Max Current(A): 0

Flux-Weakening: 0 KI: 0

☐ Automatic logout enablers

Logout time(S): 0

Recovery time(S): 0

KI setting:

Adjusting motor high-speed jitter

1.Set the higher value to lower down the jitter.

2. If the motor is not jitter, lower down the value to the proper one. Range: 100-1200, 50 once time。



Automatic logout enablers :

Logout time: for S mode function to automatically reduce the speed and current when the set logout time is reached which can protect the battery.

Recovery time: the gap time for second S mode function which in order to avoid the vehicel operating in S mode for long time.

Parameter setting description

3.2 Parameter page 2

Constant speed/HHC

☐ HHC Enable

☐ Constant speed

Max. limited RPM: 0

Speed limit setting

☐ Speed limited enable

Speed ratio(%): 0

Enable or disable the speed limiting function of the controller and speed ratio settings

Limit speed= Limit ratio %* Max speed

MTPA 0

Enable or disable the slope stop function.

The braking time provided by the parking assist varies according to the vehicle torque and slope.

Constant speed and Max. limited RPM

Constant speed which means keeping motor speed constant within maximum power range

Load characteristic:

- (1) When the load increases and the speed remains the same, the current will increase;
- (2) When the load decreases and the speed remains the same, the current will decrease.

Flux weakening with max. limited RPM: the max. limited speed should be set equal to the motor landing speed. If the setting value is lower than the motor rated speed, the flux weakening value can be increased or stay at same. Otherwise, it has to be increased.

Notice: Due to the loss of the motor on the ground relative to no load, the maximum speed limit setting value is usually $\leq$ (motor rated speed + flux weakening value) RPM.

To determine whether the highest speed flux weakening value is reasonable, first check the maximum speed limit setting value under the speed of the D-axis current, according to the D-axis current value multiplied by 1.2 times is the highest weak magnetic value of the setting value (Flux weakening in Page 2)

For example: the maximum speed limit value of 800, the motor rotates to 800 revolutions, the D-axis current is shown to be 500, then the highest speed weak magnetic value: $500 \times 1.2 = 600$

MTPA: For maximum torque adjustment of mid drive motor with V-magnets

To find the best match between current and torque so that the motor can produce maximum torque with minimum current. (Multiple tests of the motor are required)

Hub motor : 15 (Fixed)

Mid drive motor: below 2000W= 45、2000W=65、3000W=85、above 3000W=105。

Parameter setting description

3.2 Parameter page 2

Max three-speed RPM settings:

Low speed current=Low-speed ratio%*Maximum limited RPM

Mid speed current=Mid-speed ratio%*Maximum limited RPM

High speed current=High-speed ratio%*Maximum limited RPM

Starting phase current: Maximum phase current limit value corresponding to the first acceleration section

Starting phase current ≤ controller maximum phase current value setting

KP:

Adjust the sound and resonance of the motor when starting. The louder the motor sound as well as the resonance, the smaller the value is. Otherwise, increase the value. And taking a reasonable value.

Three speed default gear:

Selecting the power-on default gear

Button 3 speed is effective

Three-speed	
Low(%): 0	Low speed current(%): 0
Mid(%): 0	Mid speed current(%): 0
Hige(%): 0	Over Modulation: 0
Start phase current: 0	protect time: 0
KP: 0	Limit phase current: 0
Button/Switch 3 speed	
☒ Button 3 speed ☐ Switch 3 speed	
Three speed default gear	Soft start enabled
☒ Low ☐ Mid ☐ Hige	Soft start grade: 1

Soft start:

Unchecked for hard start.
The higher the level the more power when starting.

Three-speed main bus current settings:

Low-speed current=Low-speed ratio%*Maximum main bus current

Mid-speed current=Mid-speed ratio%*Maximum main bus current

Over modulation:

Reduce motor no-load current by increasing motor rated speed.

Hub motor: 0-14 ; Mid drive motor:10-20

Phase current settings:

Protect the controller by setting the maximum phase current maintenance time through the capacitor protection time.

(The protection time and limiting phase current parameters are filled in according to the documentation of the Technical Department.)

Button/Switch 3 speed :

Button 3 speed: press to select the gear

Switch 3 speed: switch to select the gear

(Port settings should be consistent with the way here)



Parameter setting description

3.2 Parameter page 3

Hall, Phase wire exchange: To decide the motor rotate direction. Forward or backward

Motor Setting

Pole pairs:
4

☐ exchange hall wire color Yellow-Blue
☐ exchange phase wire color Blue-Green

Motor type
☒ surface-mount ☐ V-type

Hall shift Angle
0

Motor Pole Pairs: According to the motor
(Wrong value will affect the motor speed)

Different motor types have different pole pairs

Hall shift Angle: According to motor's spec
(For hub motor, normally , it is -60)

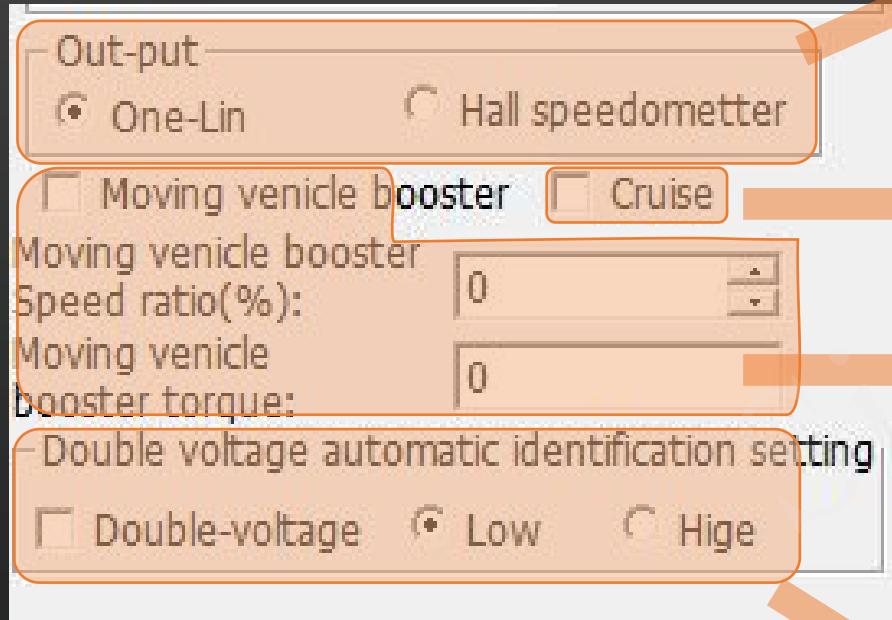


The rotor magnet fixing type of motor:
For the hub motor, choose the surface-mount (the position corresponding to the Hall).

For mid drive motor, tick V-type

Parameter setting description

3.2 Parameter page 3



The screenshot shows a software interface for parameter settings. It includes several sections: 'Out-put' with radio buttons for 'One-Lin' (selected) and 'Hall speedometer'; 'Moving venide booster' with checkboxes for 'Moving venide booster' and 'Cruise'; two numeric input fields for 'Moving venide booster Speed ratio(%)' and 'Moving venide booster torque', both set to 0; and 'Double voltage automatic identification setting' with checkboxes for 'Double-voltage' and radio buttons for 'Low' (selected) and 'Hige'.

Speedmeter communication methods: One-line communication or Hall Speedmeter.

(The port settings: One-line communication or Hall signal it should be unified here)

Cruise function On or Off .

(Port setting needs to set Cruise function)

Moving vehicle booster Function On or Off:

Used for 2 wheelers

Moving vehicle booster Function=Speed ratio% * max speed limit

Moving vehicle booster Torque=max phase current

Pls note: Speed 3km/h, torque 9~19N.m

Double voltage Setting: Single voltage as default, low voltage as default

Pls note: The speed value for double voltage function is set according to motor rpm (Port setting: Need to choose double voltage function)

Parameter setting description

3.2 Parameter page 3

Reversing the speed limit(%): 0

EBS ratio(%): 0

☐ Low beake ☐ Secure boot

Reverse Speed limit Setting: Reverse Speed = Reverse limited speed%*max limited speed

EBS ratio: Set electric brake regen power
Set parameter between 8~20 for 2 wheeler
Set parameter between 20~30 for 3 or 4 wheeler
(Setting according to the comfort feeling when release the handlebars and brake)

Secure boot Function On or Off: Off as default (Special soft function)

Low brake enable: brake signal high and low brake, high brake as default, check low brake

Parameter setting description

3.4 Setting for advanced page

LD-EM-V3

PAGE1 PAGE2 PAGE3 Display Port Settings

Motor Setting

Pole pairs: 4

☐ exchange hall wire color Yellow-Blue
☐ exchange phase wire color Blue-Green

Motor type
☒ surface-mount ☐ V-type

Hall shift Angle: 0

Out-put
☒ One-Lin ☐ Hall speedometer

☐ Moving vehicle booster ☐ Cruise

Moving vehicle booster
Speed ratio(%): 0

Moving vehicle
booster torque: 0

Double voltage automatic identification setting
☐ Double-voltage ☒ Low ☐ Hige

Reversing the speed limit(%): 0

EBS ratio(%): 0

☐ Low beake ☐ Secure boot

PORT: COM1 OPEN

Status: research

Connect

success_count: 0

error_count: 0

import prarm

save prarm

Param write



communicatin and download set

set compare check 585

current check 585

remote control

Throttle: 0.50 V

Gear
☒ L ☐ M ☐ H ☐ S ☐ Switch

☐ Brake ☐ R ☐ Lock

Calibration

vol cal: 1490 CLR DTC

cur cal: 81

weak flux: 15

open file

download

moden: No-ops delay: 12

Parameter setting description

3.4 Setting for advanced page

The screenshot shows a software interface for parameter settings. Orange boxes highlight the following sections:

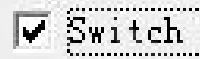
- Download process display:** A box at the top containing 'set compare check' (value 585) and 'current check' (value 585).
- Remote control interface:** A box on the left containing 'Throttle' (0.58 V), 'Gear' (L, M, H, S), and 'Brake', 'R', 'Lock' checkboxes.
- CLR DTC:** A button next to the 'Calibration' section.
- Calibration:** A section containing 'vol cal' (1490), 'cur cal' (81), and 'weak flux' (15).
- Open file/Download/Mode:** A box at the bottom containing 'open file', 'download', 'moden' (No-ops), and 'delay' (12).

Arrows point from these highlighted sections to descriptive text blocks on the right.

Download process display

1. The actual value corresponds to the first inspection result of the controller parameter software.
2. The comparison value compares to the controller from same batch. If the two values are same, then parameters are consistent.

Used to switch the display interface. If not selected, the speed/voltage/current will be displayed.



Display RPM/Q axle voltage/D axle voltage



Display RPM/Q axle current/D axle current

CLR DTC

Voltage calibration: Adjust whether the displayed voltage is consistent with the actual voltage. No need to change.
Current calibration: Adjust whether the displayed current is consistent with the actual current. No need to change.
Weak Flux: Adjust D-axls voltage and Q-axls current.

Open file: Open ECU bin software

Download: Import the ECU program file or configuration file into the controller

Mode: No need to choose normally

Remote control interface

Parameter setting description

3.5 Setting for Display page

Display motor speed
RPM in real time

Display controller Voltage
in real time

Display controller current
in real time

LD-EM-V3

PAGE1 PAGE2 PAGE3 Display Port Settings

RPM: 0

Voltage: 0

Current: 0

Controller Status:

Fault Code: IDLE

Fault display

- ☐ Hall Fault
- ☐ Brake
- ☐ Motor hall
- ☐ Undervoltage
- ☐ Overvoltage
- ☐ Overcurrent
- ☐ Controller Failure
- ☐ Out of control
- ☐ Overheating

Controller temp(?)

External temp(?)

Temp coefficient

Gears

- ☒ L ☐ M
- ☐ H ☐ S
- ☒ R ☒ P
- ☒ Brake ☒ ANT
- ☒ SIDE ☒ Regi
- STAND

PORT: COM1 OPEN

Status: research

Connect

Param write

success_count: 0

error_count: 0

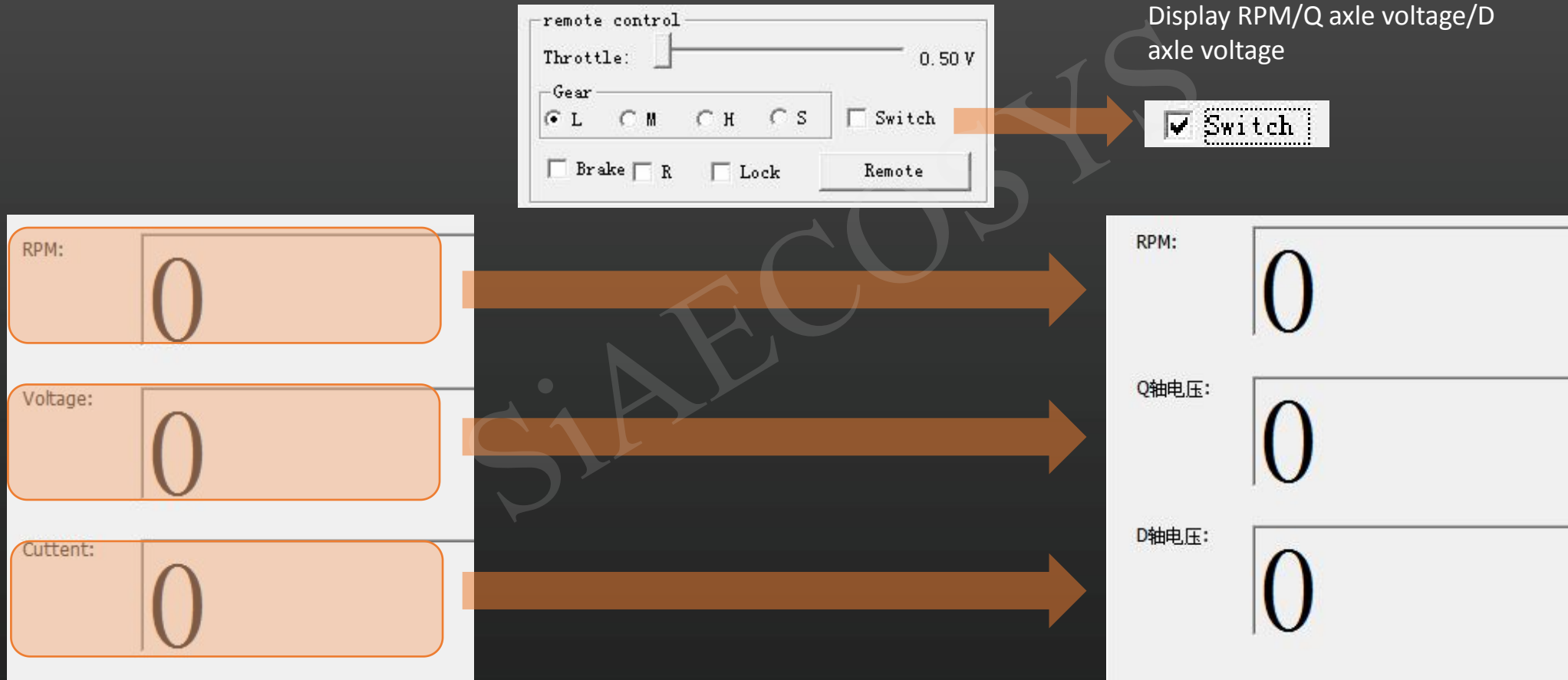
import prarm

save prarm

Display fault in real time

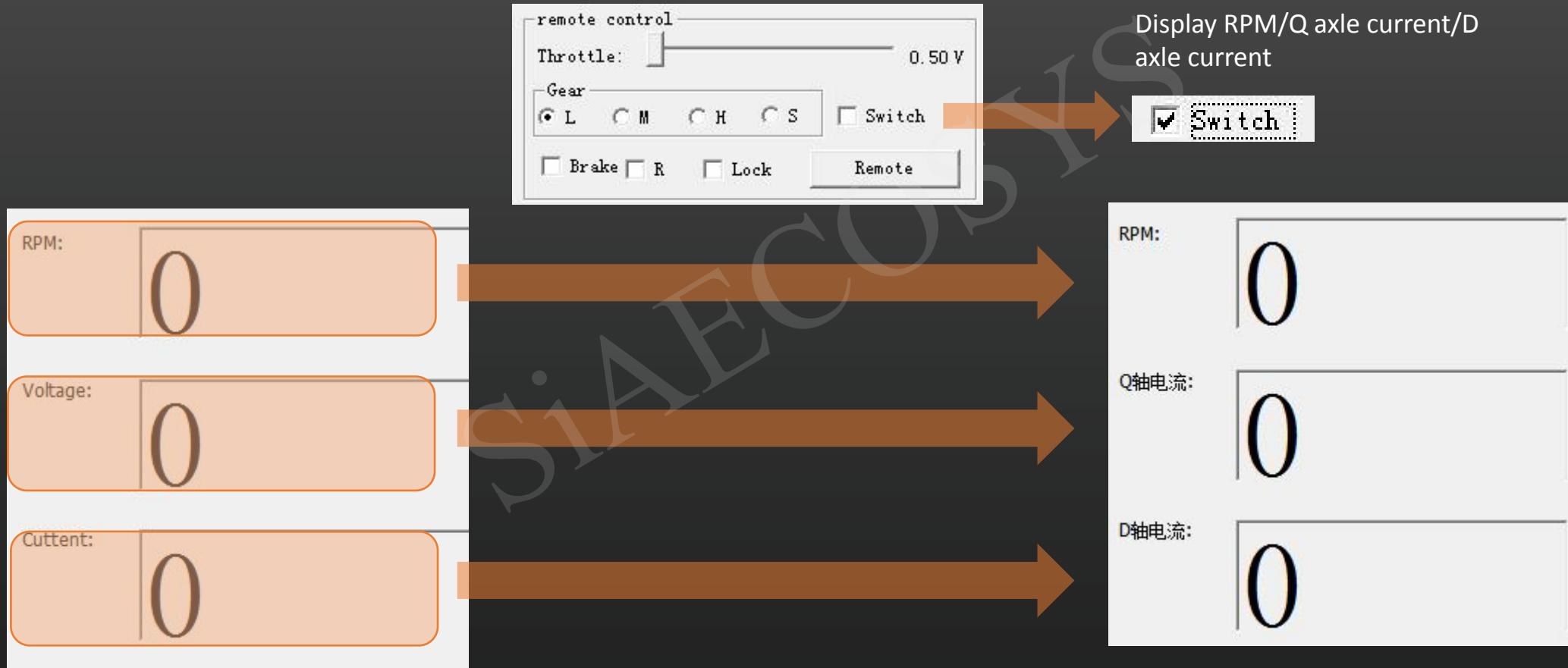
Parameter setting description

3.5 Setting for Display page



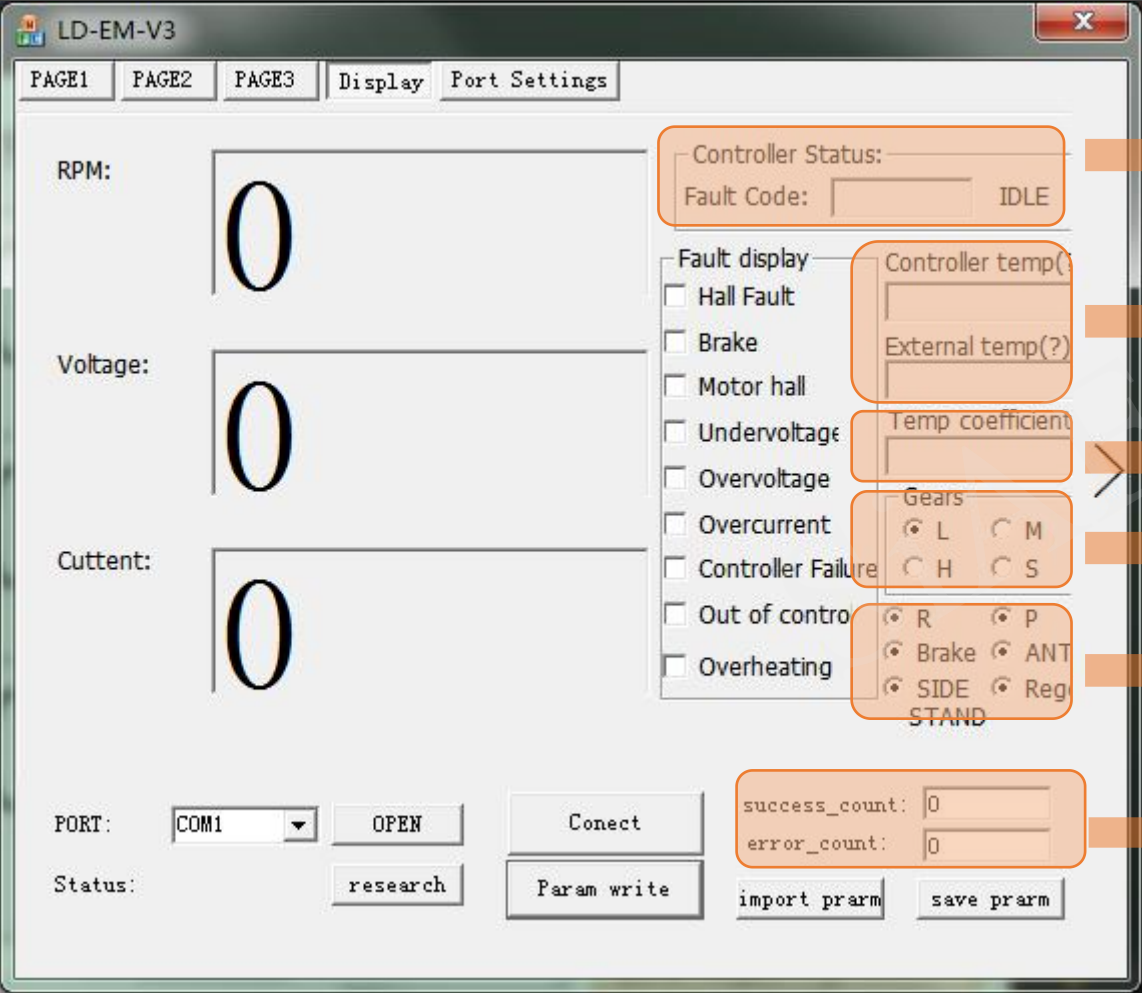
Parameter setting description

3.5 Setting for Display page



Parameter setting description

3.5 Setting for Display page



Fault code: Display fault code

Controller temp: Display controller temp

External temp: Display motor temp

Temp coefficient: Display the controller hardware value

Gears: Display the gear in real time

Status: Display the controller status in real time

Connection: Display if controller connection is normal

Parameter setting description

3.6 Setting for Advanced page

LD-EM-V3

PAGE1 PAGE2 PAGE3 Display Port Settings

Motor Setting

Pole pairs: 5

☒ exchange hall wire color Yellow-Blue

☒ exchange phase wire color Blue-Green

Motor type

☐ surface-mount ☒ V-type

Hall shift Angle: 176

Reversing the speed limit(%): 8

EBS ratio(%): 20

☐ Low beake ☐ Secure boot

Out-put

☐ One-Lin ☒ Hall speedometer

☐ Moving vehicle booster ☐ Cruise

Moving vehicle booster

Speed ratio(%): 10

Moving vehicle booster torque: 1000

Double voltage automatic identification setting

☐ Double-voltage ☒ Low ☐ Hige

PORT: COM1 OPEN

Status: research

Connect

Param write

success_count: 0

error_count: 0

import prarm

save prarm

In any page, error_count enters ~ and then click shift button on the keyboard, then you can open the remote control function in extended page

success_count: 0

error_count: ~

Parameter setting description

3.6 parameter description (Advanced interface)

The screenshot shows a software interface for motor parameter settings. It is divided into several sections: 'Motor Setting', 'Reversing the speed limit(%)', 'EBS ratio(%)', 'Motor noise reduction', 'Secure boot', 'Temperature protection', 'Out-put', 'Moving vehicle booster', and 'Double voltage automatic identification setting'. The 'Temperature protection' section is highlighted with an orange border and contains five rows of settings: 'Initial temperature' (120), 'Over-temperature' (140), 'Max temperature' (150), 'T.C.1:' (105), and 'DAMPING' (10). The 'Display speed adjustment:' row is also highlighted and contains the value 9428. Arrows point from the text descriptions on the right to the corresponding input fields in the 'Temperature protection' section.

Parameter	Value
Pole pairs	24
Reversing the speed limit(%)	6
EBS ratio(%)	15
Motor type	surface-moun
Hall shift Angle	60
Out-put	One-Lin
Moving vehicle booster Speed	10
Moving vehicle booster torque	320
Initial temperature	120
Over-temperature	140
Max temperature	150
T.C.1:	105
DAMPING	10
Display speed adjustment:	9428

Motor temp protection setting (Port settings: The corresponding motor temperature sensor model needs to be configured)

Initial temperature: the controller enters temperature protection to limit power when reaches the set temperature.

Over temperature: the controller enters the second stage temperature protection to limit the power when reaches the set temperature.

Max temperature: the controller cuts off the motor output when reaches the set temperature.

Controller temp protection settings (fill in the corresponding parameters according to the documents issued by the technical department)

The controller enters the protection temperature: After reaching the set temperature, the controller enters the temperature protection and limits the power.

The controller stops output when reaching the set temperature +20°C

DAMPING (range: 10~30)

Prevent motor auto-rotation failure. If the value is filled in too much, the motor will stop too fast and a current sound will appear when it is stationary.

used to adjust speedometer display speed

Fill in the multiple of 265

Parameter setting description

3.6 parameter description (Advanced interface)

The screenshot shows a software interface for parameter settings. It includes fields for 'set compare check' (558) and 'current check'. Under 'remote control', there is a 'Throttle' slider set to 0.50V, a 'Gear' section with radio buttons for L, M, H, and S, and checkboxes for 'Brake', 'R', 'Lock', and 'Remote'. A 'Calibration' section contains 'vol cal' (1494), 'cur cal' (119), and 'weak flux' (30), along with a 'CLR DTC' button. At the bottom, there are 'open file' and 'download' buttons, a 'moden' dropdown set to 'No-ops', and a 'delay' field set to 12. A 'RESET' button and a 'CAN' checkbox are also present.

Analog throttle voltage controls motor operation

Analog gear, used to gear-switch

Remote control and local control switching

Analog vehicle switch

CAN mode

Adjust CAN debugging line baud rate and ID
Control debug line restart

Used for writing
calibration data

Used to restart the controller
to clear faults

This detailed view of the 'Calibration' section shows the following controls:

- 'vol cal': 1494
- 'cur cal': 119
- 'weak flux': 30
- 'CLR DTC' button
- 'Auto_B' checkbox (unchecked)
- Baud rate selection: 1M (selected), 500, 250, 12E
- ID selection: A (selected), B, C, D, E, F, G, H
- 'Calibration' button
- '重启调试线' (Restart debug line) button
- 'RESET' button
- 'CAN' checkbox (checked)

Parameter setting description

3.7 parameter description (port interface)

The screenshot displays a parameter setting interface for a port interface, organized into a 4x4 grid of 16 individual port configuration blocks. Each block is labeled with a port name and contains the following controls:

- Three checkboxes: **IO**, **SW**, and **LA**.
- Three radio buttons: **F**, **U**, and **D**.
- A dropdown menu labeled **1:empty_func**.

The ports shown in the grid are: PD0, JTCK, SWD, PA11, PB3, PD1, PA12, PC15, PA0, PB9, PB4, PA15, PB2, PC14, PB5, and PD15(empty). In all blocks, the **IO** checkbox is checked, and the **F** radio button is selected. The dropdown menu in each block is set to **1:empty_func**.

The port interface selects port functions according to the actual needs of the vehicle.

The port and the actual corresponding location need to be set according to the (port description diagram of the document issued by the technical department)

IO is output

SW is for jog

LA is an effective switch between negative control and positive control.

F U D is the function branch (for example, for the one-line communication function, select F for the Hall number output and select U for the speed output)

PD15 defaults to option 19. The CAN controller can select CANID through this port controller.

Parameter setting description

3.7 parameter description (port interface)

The port function and the setting interface function selection must be consistent.

Example: EM80 port switch 3 speed settings

PD0 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func	JTCK ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func	SWD ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 6:high_speed_sv	PA11 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 10:reverse_set
PB3 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 18:sport_input	PD1 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func	PA12 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 8:stealavoid	PC15 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 9:lamhome
PA0 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func	PB9 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 11:single_wire_cr	PB4 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 12:side_sustain	PA15 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func
PB2 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 2:low_speed_swi	PC14 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 22:LY_e_gear	PB5 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 16:high_break	PD15(empty) ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 19:undef_IO_op

Current/Flux-Weakening Max Current(A): 80 Flux-Weakening: 3000 KI: 220 ☐ Automatic logout enadlers Logout time(S): 30 Recovery time(S): 30	MTPA 15 Three-speed Low(%): 60 Low speed current(%) 80 Mid(%): 80 Mid speed current(%) 100 Hige(%): 100 Over Modulation: 5 Start phase current: 6000 protect time: 0 KP: 9000 Limit phase current: 6500
Constant speed/HHC ☐ HHC Enable ☒ Constant speed Max. limited RPM: 1000	Button/Switch 3 speed ☐ Button 3 speed ☒ Switch 3 speed
Speed limit setting ☐ Speed limited enable Speed ratio(%): 42	Three speed default gear ☐ Low ☒ Mid ☐ Hige ☒ Soft start enabled Soft start grade: 12

Parameter setting description

3.7 parameter description (port interface)

The port function and the setting interface function selection must be consistent.

Example: EM80 port switch 3 speed settings

PD0 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func	JTCK ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func	SWD ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 6:high_speed_sv	PA11 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 10:reverse_set
PB3 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 18:sport_input	PD1 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func	PA12 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 8:stealavoid	PC15 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 9:lamhome
PA0 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func	PB9 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 11:single_wire_cr	PB4 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 12:side_sustain	PA15 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func
PB2 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 2:low_speed_swi	PC14 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 22:LY_e_gear	PB5 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 16:high_break	PD15(empty) ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 19:undef_IO_op

Current/Flux-Weakening Max Current(A): 80 Flux-Weakening: 3000 KI: 220 ☐ Automatic logout enadlers Logout time(S): 30 Recovery time(S): 30	MTPA 15 Three-speed Low(%): 60 Low speed current(%) 80 Mid(%): 80 Mid speed current(%) 100 Hige(%): 100 Over Modulation: 5 Start phase current: 6000 protect time: 0 KP: 9000 Limit phase current: 6500
Constant speed/HHC ☐ HHC Enable ☒ Constant speed Max. limited RPM: 1000	Button/Switch 3 speed ☐ Button 3 speed ☒ Switch 3 speed
Speed limit setting ☐ Speed limited enable Speed ratio(%): 42	Three speed default gear ☐ Low ☒ Mid ☐ Hige ☒ Soft start enabled Soft start grade: 12

Parameter setting description

3.7 parameter description (port interface)

The port function and the setting interface function selection must be consistent.

Example: EM80 port One Line speedometer settings

PD0 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func	JTCK ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func	SWD ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 7:egear_buttan	PA11 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 10:reverse_set
PB3 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 18:sport_input	PD1 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func	PA12 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 8:stealavoid	PC15 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 9:lamehome
PA0 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func	PB9 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 11:single_wire_c	PB4 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 12:side_sustain	PA15 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func
PB2 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func	PC14 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 22:LY_e_gear	PB5 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 16:high_break	PD15(empty) ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 19:undef_IO_op

Motor Setting Pole pairs: 24 ☐ exchange hall wire color Yellow-Blue ☐ exchange phase wire color Blue-Gree Hall shift Angle 60 Motor type ☒ surface-moun ☐ V-type	Reversing the speed limit(%): 6 EBS ratio(%): 15 ☐ Motor noise reduction ☐ Secure boot
Out-put ☒ One-Lin ☐ Hall speedometer ☐ Moving venicle booster ☐ Cruise Moving venicle booster Speed 10 Moving venicle booster torque: 320 Double voltage automatic identification setting ☐ Double-voltage ☒ Low ☐ Hige	Temperature protection Initial temperature 120 Over-temperature 140 Max temperature 150 T.C.1: 105 DAMPING 10 Display speed adjustment: 9428

Parameter setting description

3.7 parameter description (port interface)

The port function and the setting interface function selection must be consistent.

Example: EM80 port hall speedometer settings

PD0 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func	JTCK ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func	SWD ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 7:egear_buttan	PA11 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 10:reverse_set
PB3 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 18:sport_input	PD1 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func	PA12 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 8:stealavoid	PC15 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 9:lamehome
PA0 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func	PB9 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 13:hall_smulat_o	PB4 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 12:side_sustain	PA15 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func
PB2 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 1:empty_func	PC14 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 22:LY_e_gear	PB5 ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 16:high_break	PD15(empty) ☐ IO ☐ SW ☐ LA ☒ F ☐ U ☐ D 19:undef_IO_op

Motor Setting Pole pairs: 24 ☐ exchange hall wire color Yellow-Blue ☐ exchange phase wire color Blue-Gree Hall shift Angle 60 Motor type ☒ surface-moun ☐ V-type Out-put ☐ One-Lin ☒ Hall speedometter ☐ Moving venide booster ☐ Cruise Moving venide booster Speed 10 Moving venide booster torque: 320 Double voltage automatic identification setting ☐ Double-voltage ☒ Low ☐ Hige	Reversing the speed limit(%): 6 EBS ratio(%): 15 ☐ Motor noise reduction ☐ Secure boot Temperature protection Initial temperature 120 Over-temperature 140 Max temperature 150 T.C.1: 105 DAMPING 10 Display speed adjustment: 9428
--	--

Error code

4.1 error code

Error Code		
E_BRAKE_ON	0x001	Brake
OVER_CURRENT	0x02	Hardware overcurrent
UNDER_VOLTAGE	0x04	Under voltage
HALL_ERROR	0x08	Hall error
OVER_VOLTAGE	0x10	Over voltage
MCU_ERROR	0x20	Controller error
MOTOR_BLOCK	0x40	Motor block error
FOOTPLATE_ERR	0x80	Throttle error
SPEED_CONTROL	0x100	Run away
WRITING_EEPROM	0x200	EEROM writing
START_UP_FAILURE	0x800	Quality inspection failure
OVERHEAT	0x1000	Controller overheat
OVER_CURRENT1	0x2000	Software overcurrent
ACCELERATE_PADAL_ERR	0x4000	Throttle failure
ICS1_ERR	0x8000	Current sensor error 1
ICS2_ERR	0x10000	Current sensor error 2
BREAK_ERR	0x20000	Brake failure
HALL_SEL_ERROR	0x40000	Hall error
MOSFET_DRIVER_FAULT	0x80000	Driver failure
MOSFET_HIGH_SHORT	0x100000	MOS tube short circuit
PHASE_OPEN	0x200000	Phase wire connection failure
PHASE_SHORT	0x400000	phase wire short circuit
MCU_CHIP_ERROR	0x800000	Controller failure
PRE_CHARGE_ERROR	0x1000000	Pre-charge failure
OVERHEAT1	0x8000000	Motor overheat
SOC_ZERO_ERROR	0x80000000	SOC 0 error

Error code

4.1 error code

Error Code		
MCU_CHIP_ERROR	0x24	Controller error
FOOTPLATE_ERR	0x12	Throttle error
OVER_CURRENT	0x42	Over current
HALL_ERROR	0x13	Hall error
MAIN_RELAY_DRIVER	0x22	Relay driver error
UNDER_VOLTAGE	0x14	Under voltage
OVERHEAT	0x21	Controller overheat
CAN_COMMUNICATION_TIMEOUT	0x23	CAN communication timeout
PRE_CHARGE_ERROR	0x25	Pre-charge error
SPI_COMMUNICATION_ERROR	0x31	Sensor communication error
ANGLE_SENSOR_ERROR	0x32	Sensor error
OVERHEAT1	0x33	Motor overheat
OVERHEAT2	0x34	Radiator overheat
MOTOR_BLOCK	0x35	Motor block error
MAIN_RELAY_FAULT	0x43	Relay error
OTHER	0x11	Other
The high level blinks quickly and the low level blinks slowly		

After-sale analysis

5.1 After-sale analysis

For after-sale analysis, please refer to this file “



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